

Computed Tomography I



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology
Hyderabad

Basic principles

Geometry and historical development

Basic principles

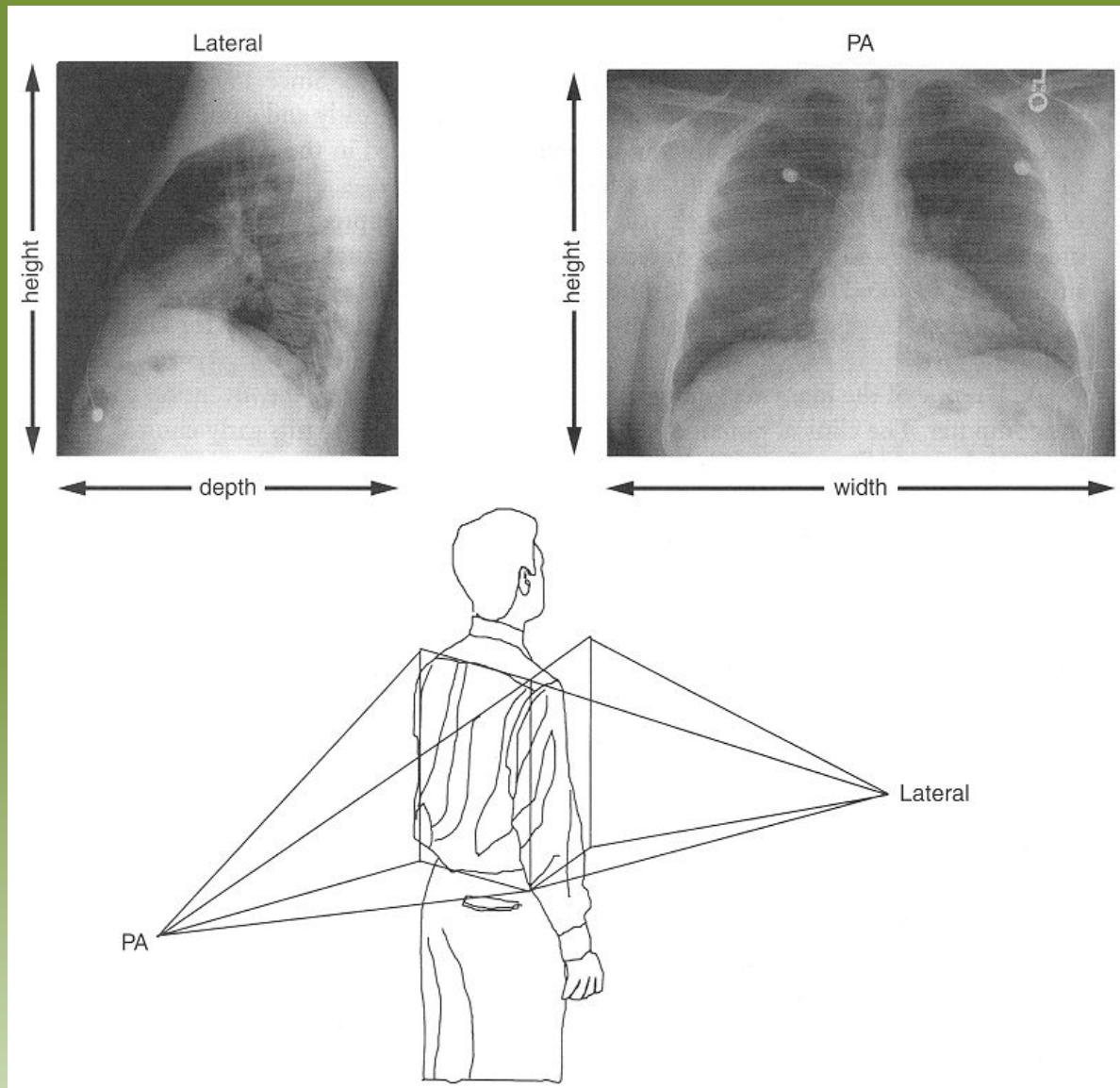
- Mathematical principles of CT were first developed in 1917 by Radon
- Proved that an image of an unknown object could be produced if one had an infinite number of projections through the object

Basic principles (cont.)

- Plain film imaging reduces the 3D patient anatomy to a 2D projection image
- Density at a given point on an image represents the x-ray attenuation properties within the patient along a line between the x-ray focal spot and the point on the detector corresponding to the point on the image

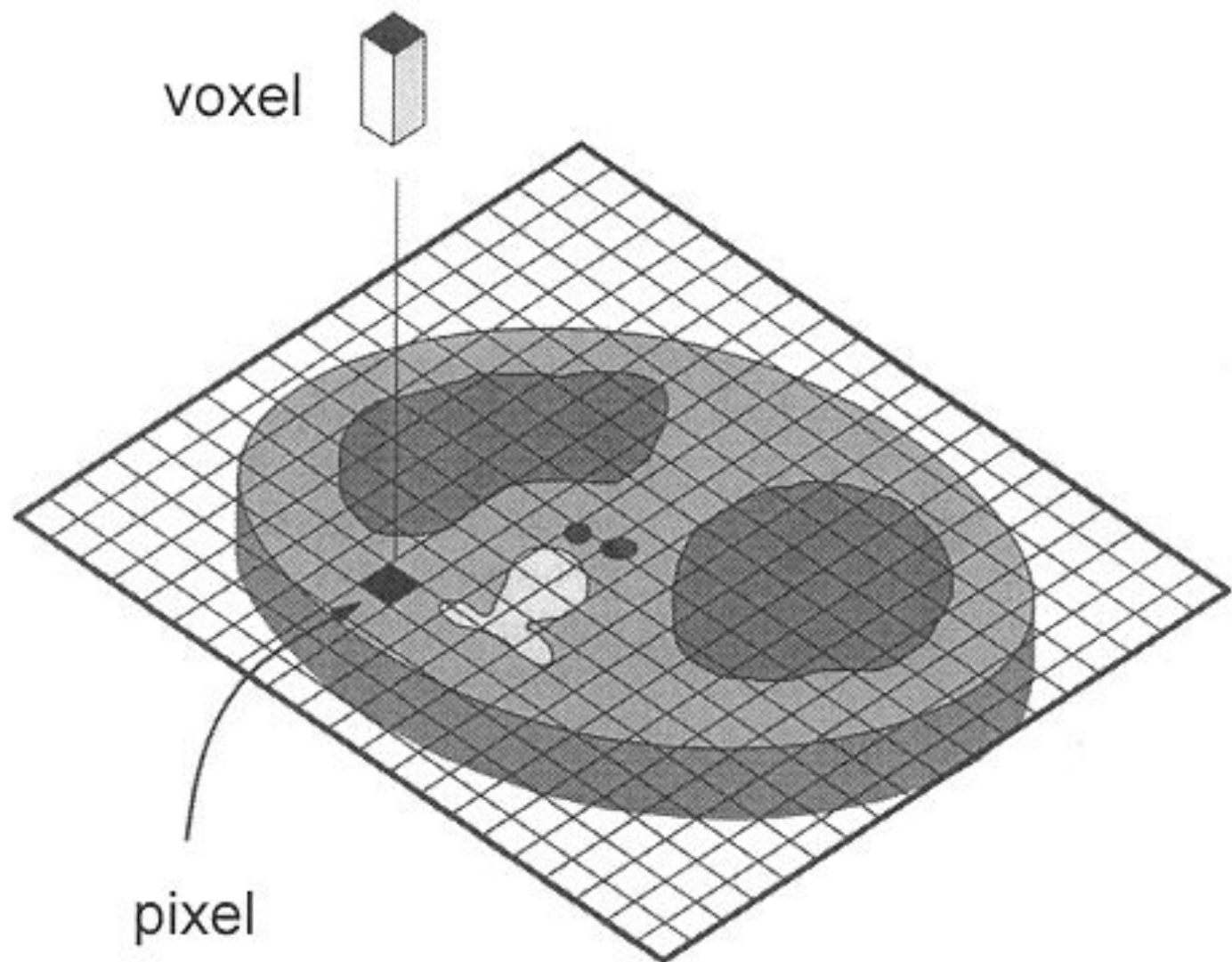
Basic principles (cont.)

- With a conventional radiograph, information with respect to the dimension parallel to the x-ray beam is lost
- Limitation can be overcome, to some degree, by acquiring two images at an angle of 90 degrees to one another
- For objects that can be identified in both images, the two films provide location information



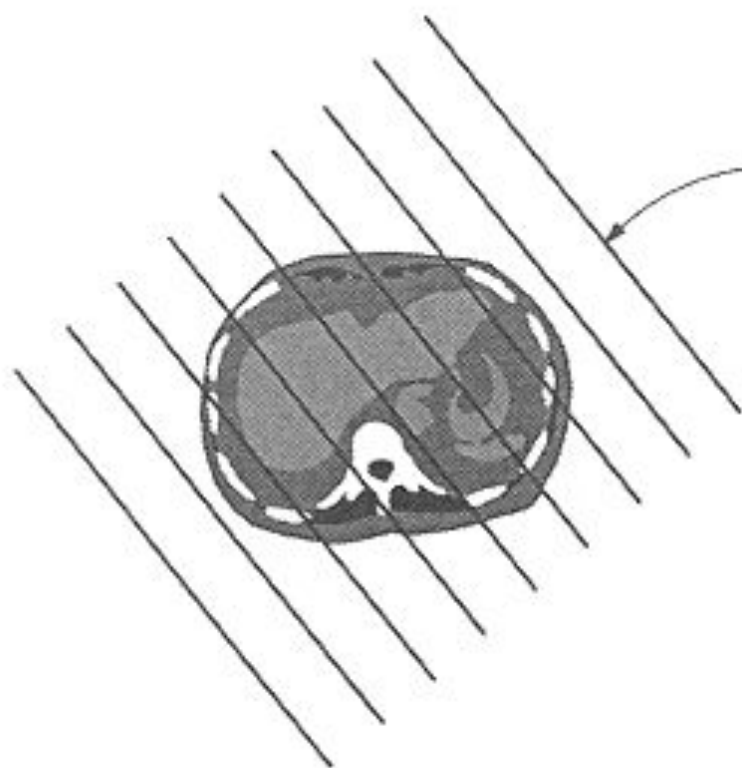
Tomographic images

- The tomographic image is a picture of a slab of the patient's anatomy
- The 2D CT image corresponds to a 3D section of the patient
- CT slice thickness is very thin (1 to 10 mm) and is approximately uniform
- The 2D array of pixels in the CT image corresponds to an equal number of 3D voxels (volume elements) in the patient
- Each pixel on the CT image displays the average x-ray attenuation properties of the tissue in the corresponding voxel

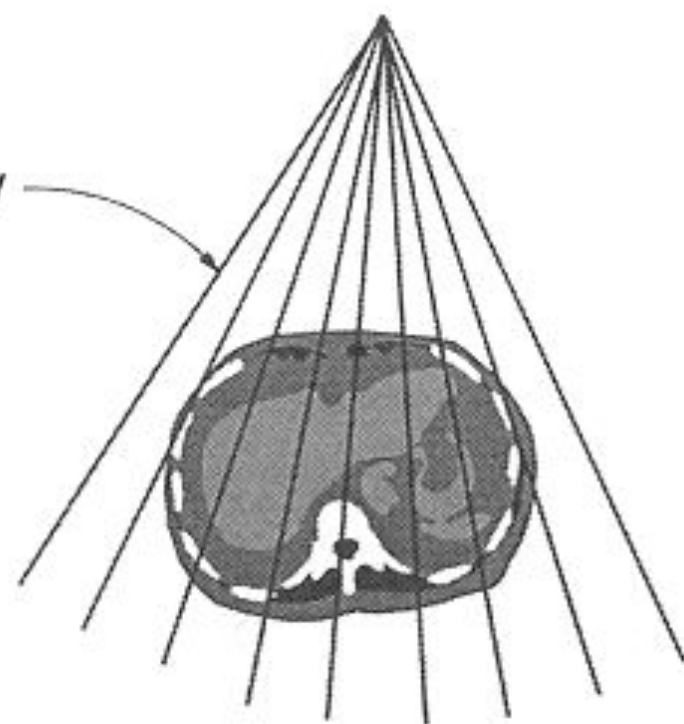


Tomographic acquisition

- Single transmission measurement through the patient made by a single detector at a given moment in time is called a *ray*
- A series of rays that pass through the patient at the same orientation is called a *projection* or *view*
- Two projection geometries have been used in CT imaging:
 - *Parallel beam geometry* with all rays in a projection parallel to one another
 - *Fan beam geometry*, in which the rays at a given projection angle diverge



parallel beam projection



fan beam projection

Acquisition (cont.)

- Purpose of CT scanner hardware is to acquire a large number of transmission measurements through the patient at different positions
- Single CT image may involve approximately 800 rays taken at 1,000 different projection angles
- Before the acquisition of the next slice, the table that the patient lies on is moved slightly in the cranial-caudal direction (the “z-axis” of the scanner)

Tomographic reconstruction

- Each ray acquired in CT is a transmission measurement through the patient along a line
- The unattenuated intensity of the x-ray beam is also measured during the scan by a reference detector

$$I_t = I_0 e^{-\mu t}$$

$$\ln(I_0 / I_t) = \mu t$$

Image Reconstruction

- Process of estimating an image from a set of projections
- Several algorithms exist to accomplish this task:
 - Brute force
 - Iterative techniques
 - Backprojection
 - Filtered backprojection

Brute force

- projection set defines a system of simultaneous linear equations - can be solved using algorithms from linear algebra
- not practical for real systems (can have hundreds of simultaneous equations for a single slice)

Reconstruction (cont.)

- *Filtered backprojection* reconstruction is most widely used in clinical CT scanners
- Builds up the CT image by essentially reversing the acquisition steps
- The μ value for each ray is smeared along this same path in the image of the patient
- As data from a large number of rays are backprojected onto the image matrix, areas of high attenuation tend to reinforce one another, as do areas of low attenuation, building up the image

The Mathematics of CT Image Reconstruction

Reconstructed image

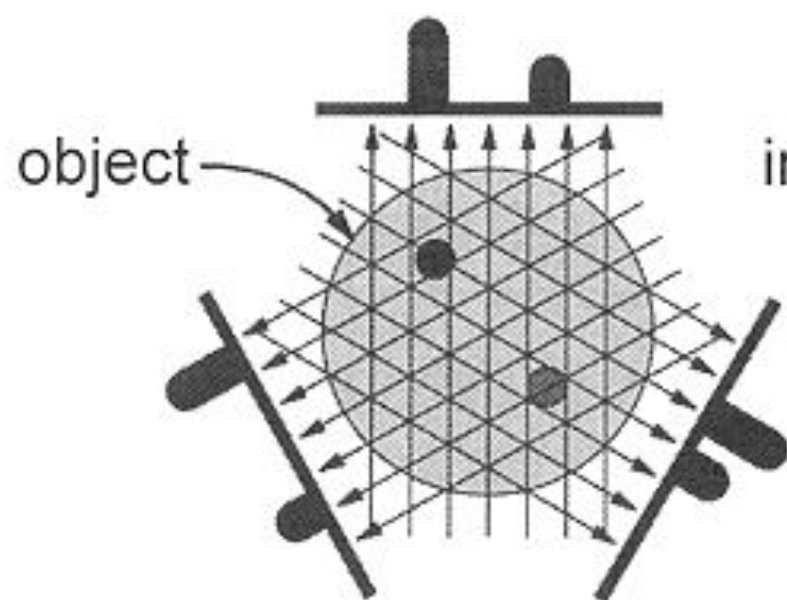
Original projections

$$g(r, \theta) = \int_0^\pi \int_{-\infty}^\infty \left[\int_{-\infty}^\infty \overbrace{f(\xi, \phi) e^{-i2\pi\rho\xi}}^{\text{Original projections}} d\xi \right] \underbrace{|\rho|}_{\text{FT of projections at each angle}} \underbrace{e^{i2\pi\rho r \cos(\theta - \phi)}}_{\text{Multiply by } |\rho|} d\rho d\phi$$

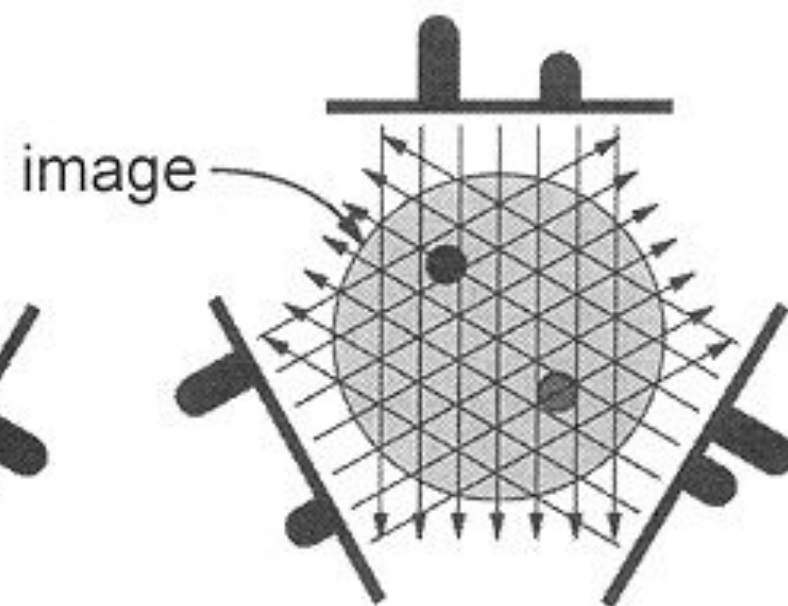
FT of projections at each angle Multiply by $|\rho|$

Invert Fourier Transform

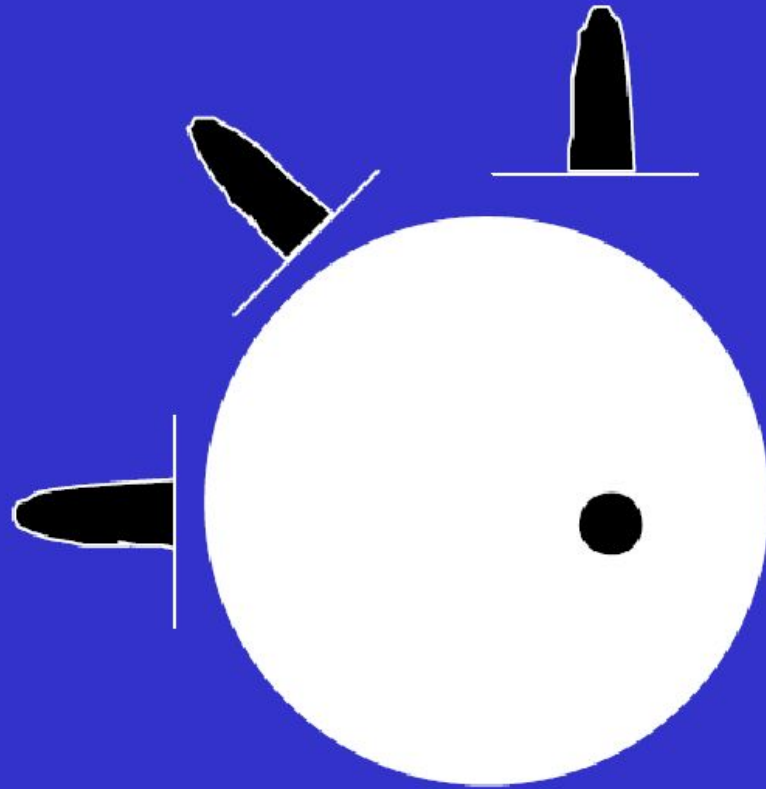
Back-project for each angle



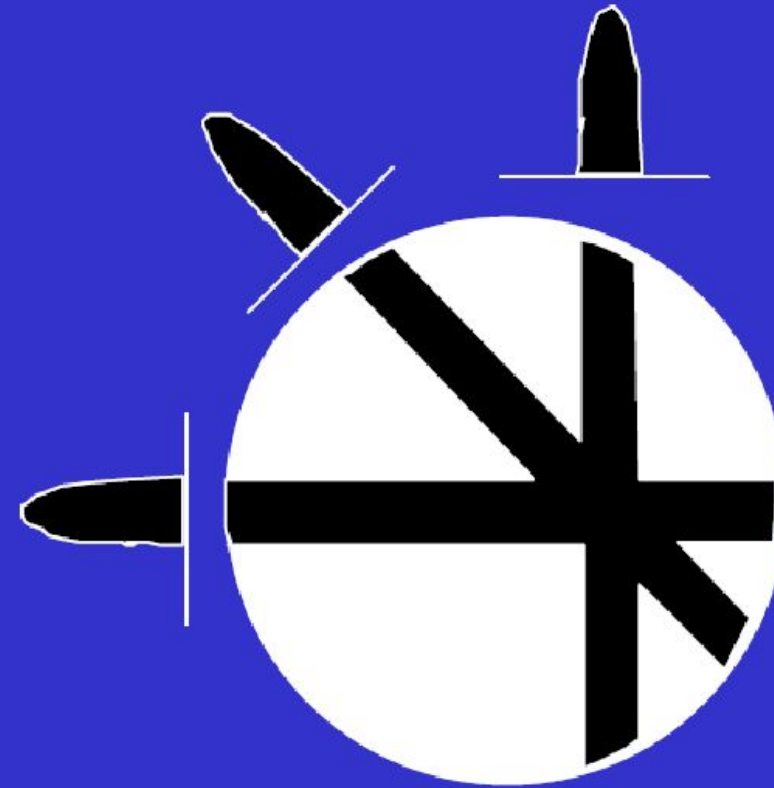
acquisition



backprojection



Projections of point object
from three directions



Back-projection onto
reconstruction plane

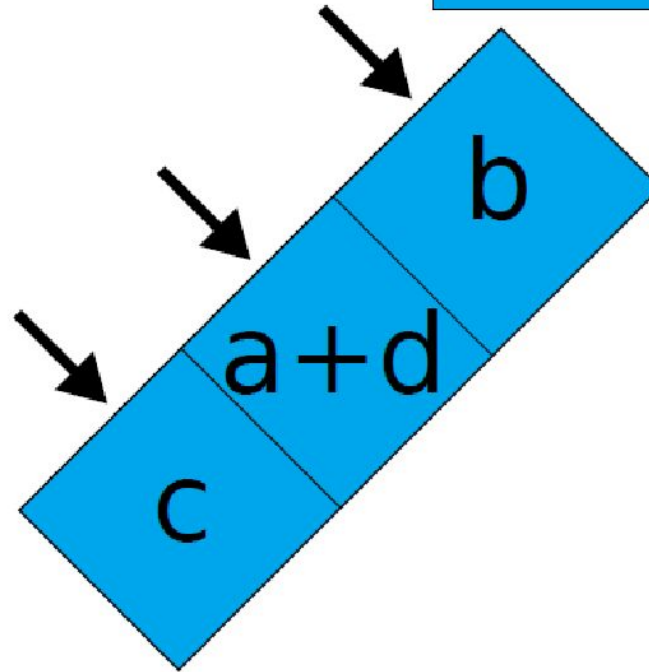
a	b
c	d



$a + b$
$c + d$

a	b
c	d

$a+b$
$c+d$



a	b
c	d

$a+b$
$c+d$

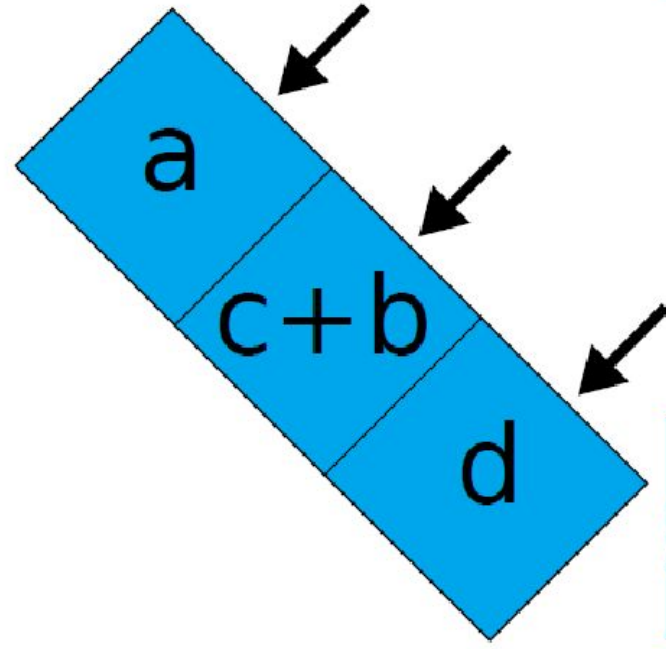
↓ ↓

$a+c$	$b+d$
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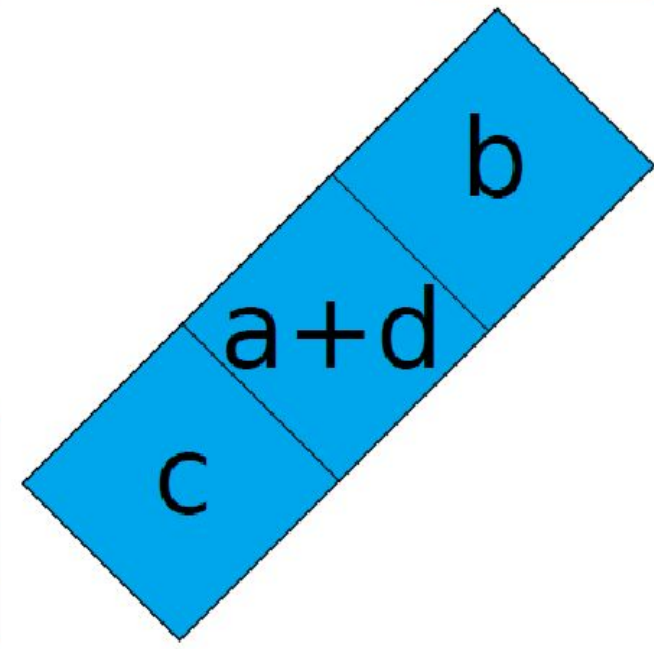
b
$a+d$
c

a	b
c	d

$a+b$
$c+d$

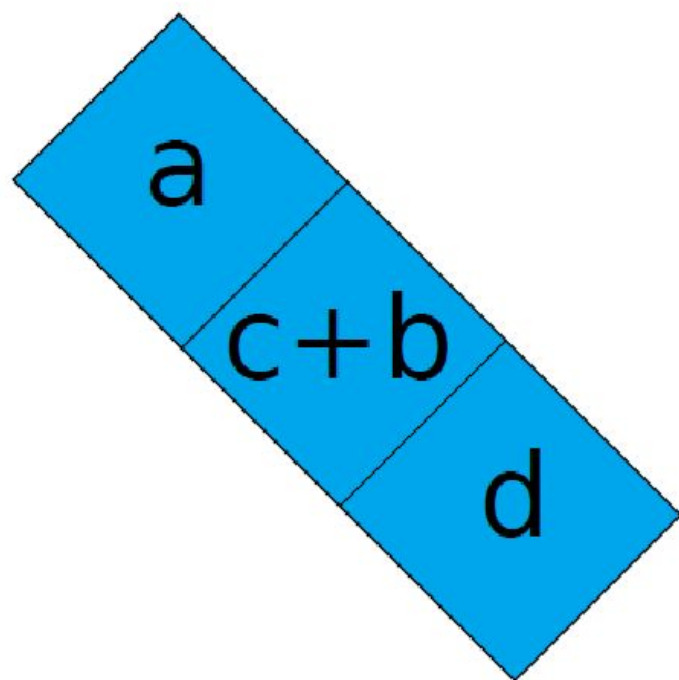


$a+c$	$b+d$
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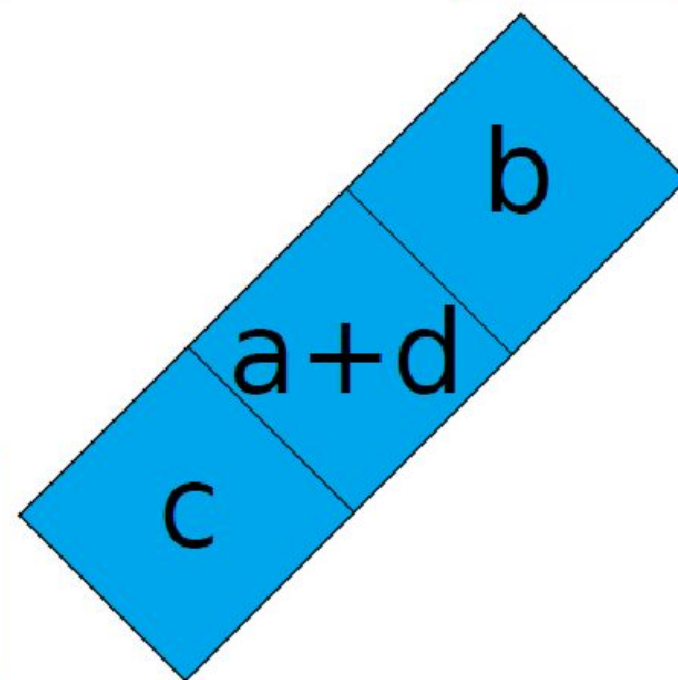


?	?
?	?

$a+b$
$c+d$

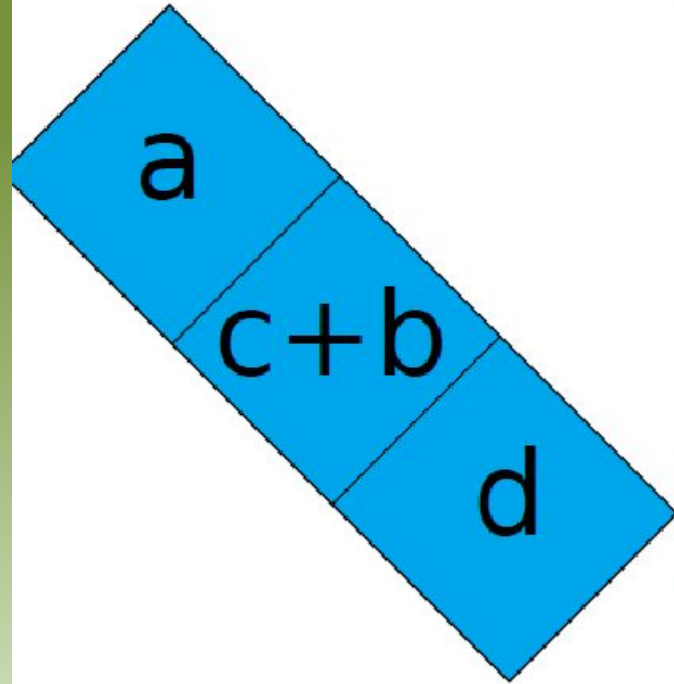


$a+c$	$b+d$
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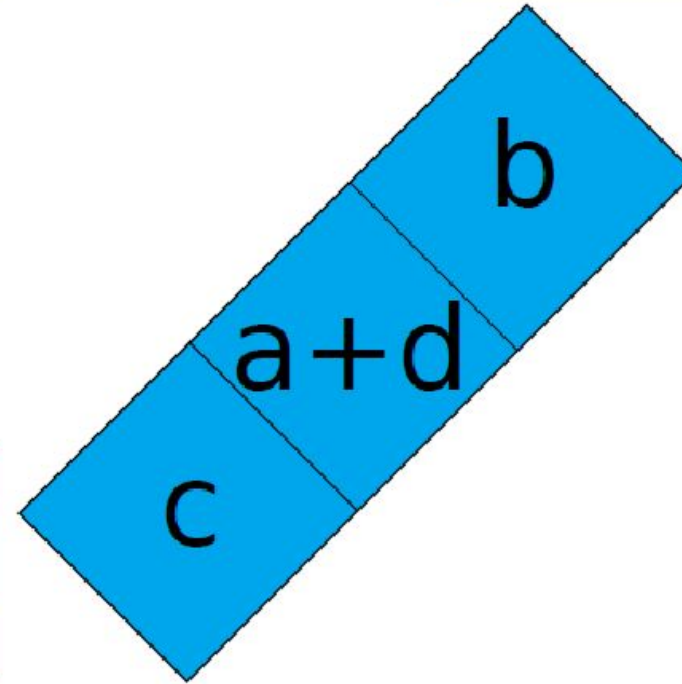


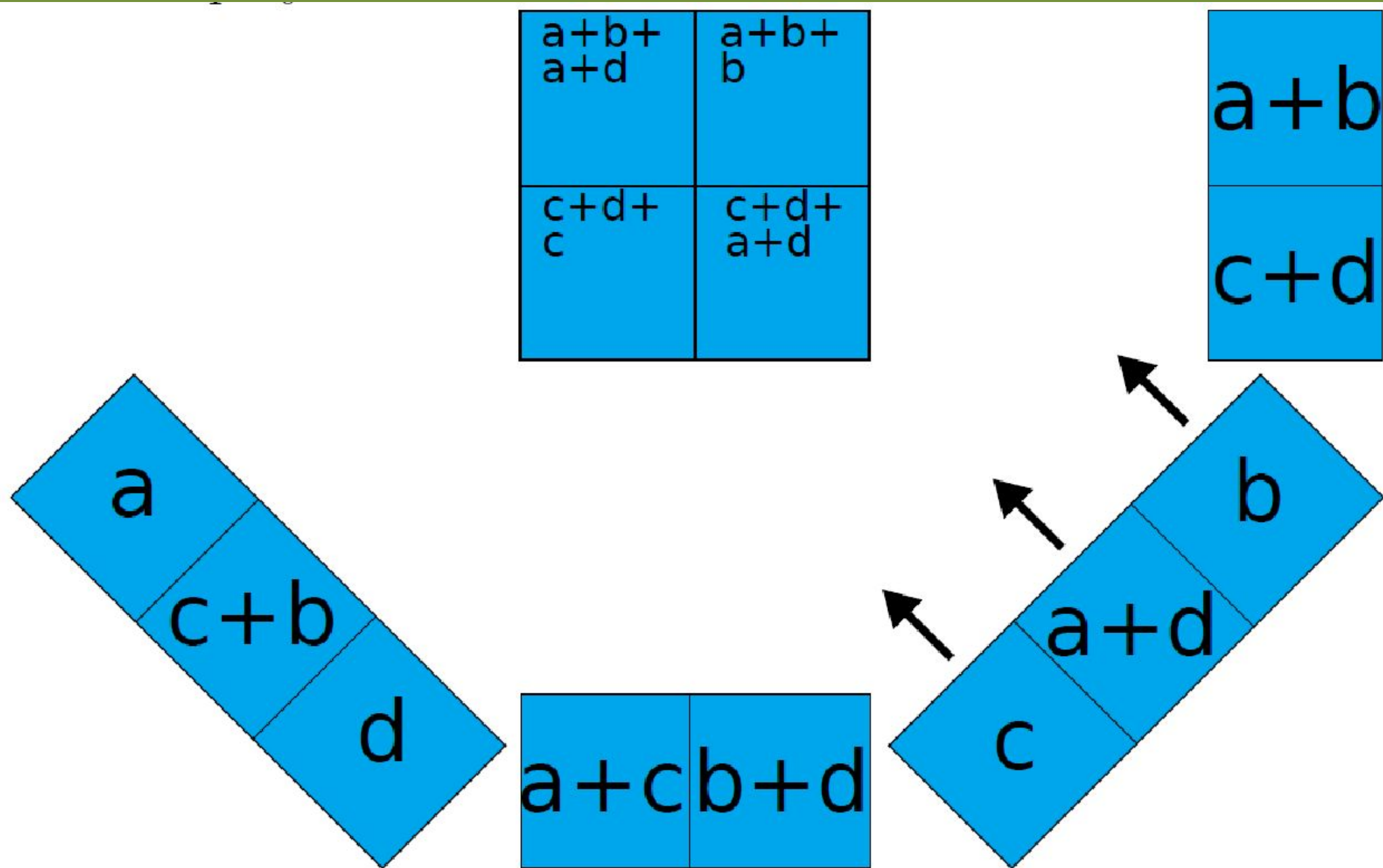
$a+b$	$a+b$
$c+d$	$c+d$

$a+b$
$c+d$



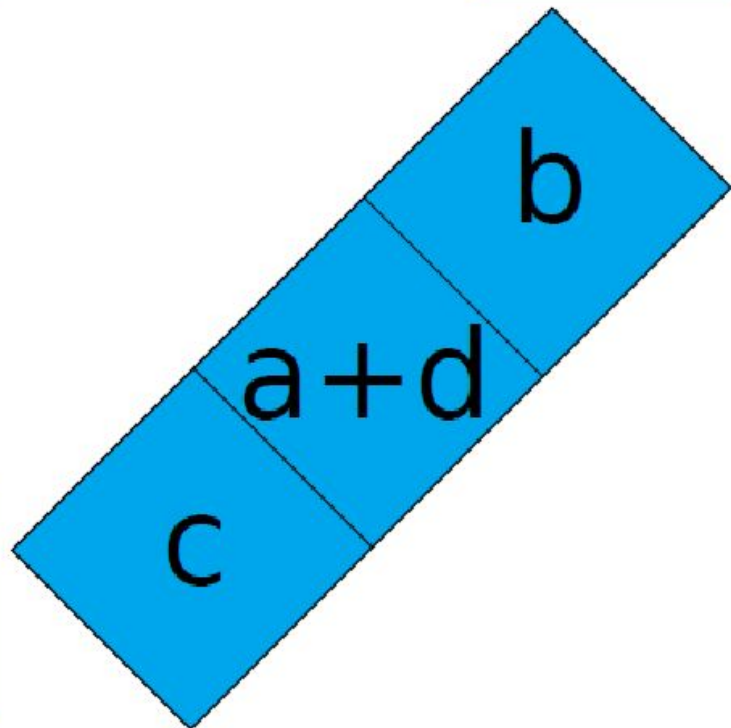
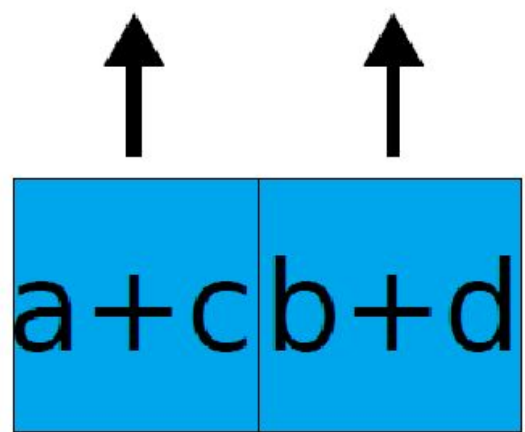
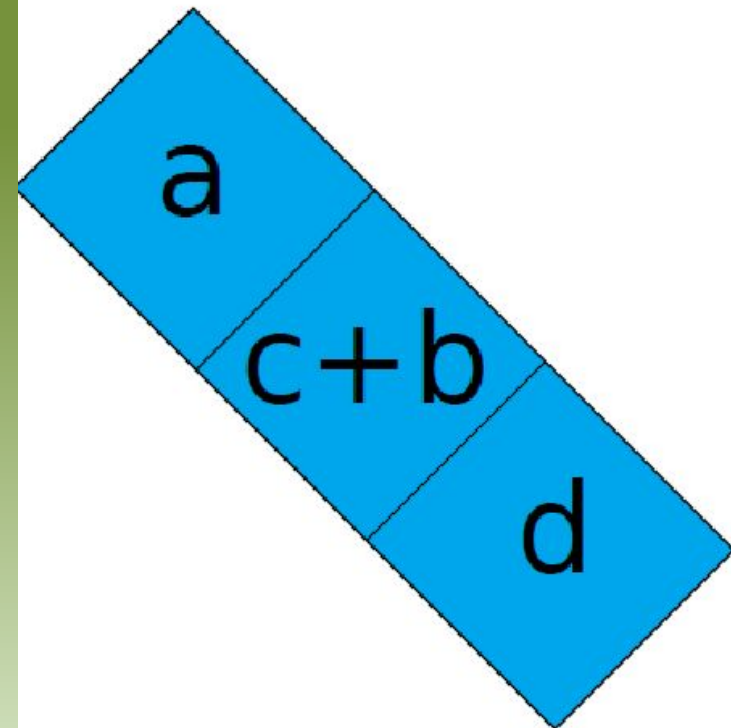
$a+c$	$b+d$
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$a+b+c$ $a+d+c$ $a+c$	$a+b+c$ $b+c$ $b+d$
$c+d+c$ $c+c$ $a+c$	$c+d+c$ $a+d+c$ $b+d$

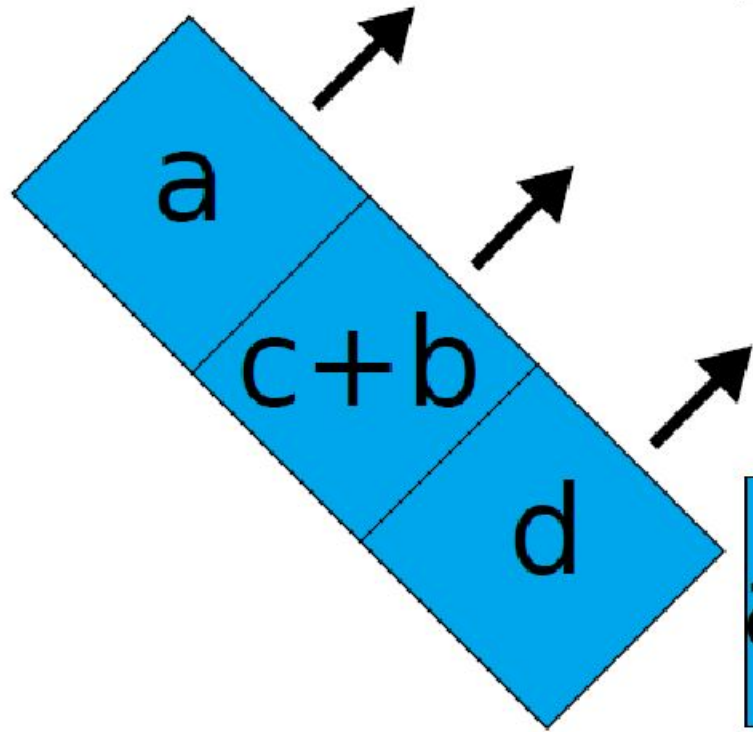
$a+b$
$c+d$



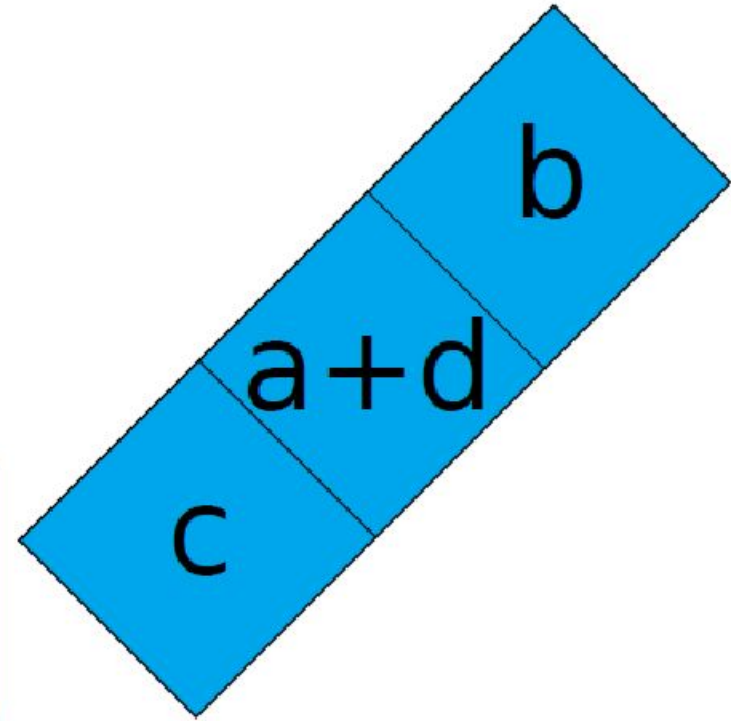
a+b+	a+b+
a+d+	b+
a+c+	b+d+
a	c+b

c+d+	c+d+
c+	a+d+
a+c+	b+d+
c+b	d

a+b
c+d



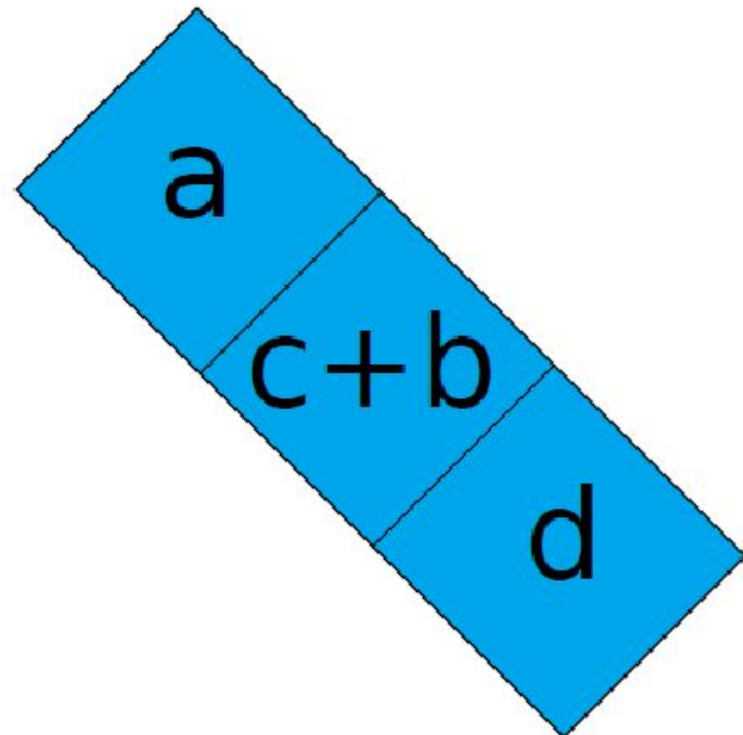
a+c	b+d
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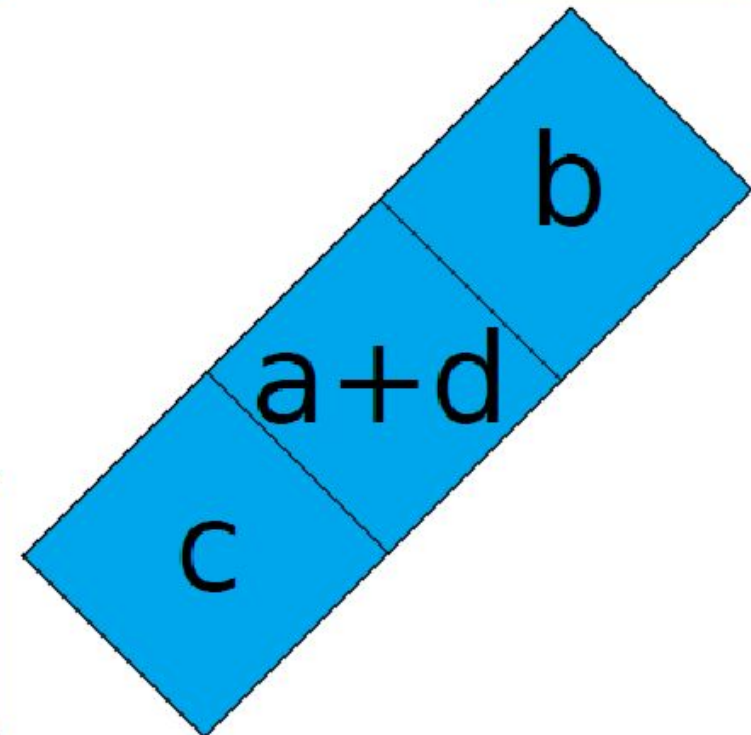
Subtract projection sum from each entry

$a+b$ $a+d$ $a+c$ a	$a+b$ b $b+d$ $c+b$
$c+d$ c $a+c$ $c+b$	$c+d$ $a+d$ $b+d$ d

$a+b$
$c+d$



$a+c$	$b+d$
-------	-------



Subtract projection sum from each entry

a a a	b b b
c c c	d d d

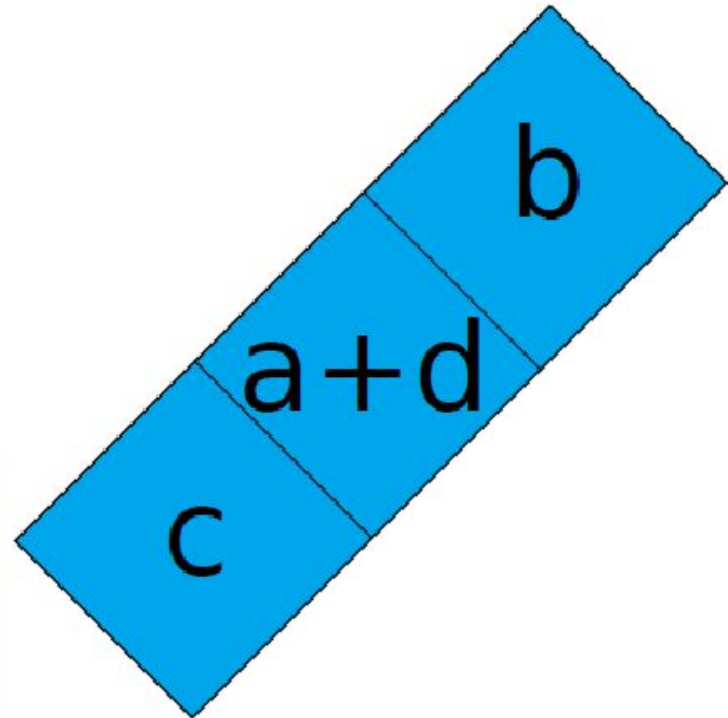
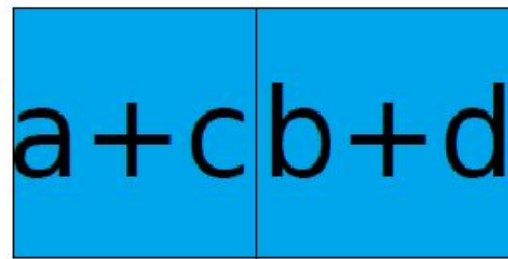
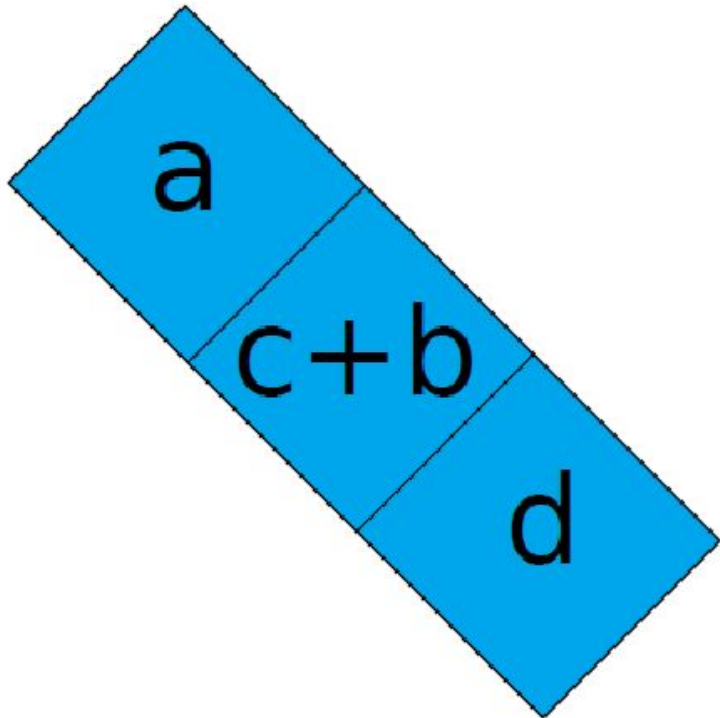
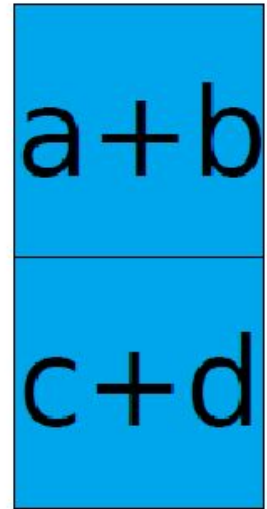
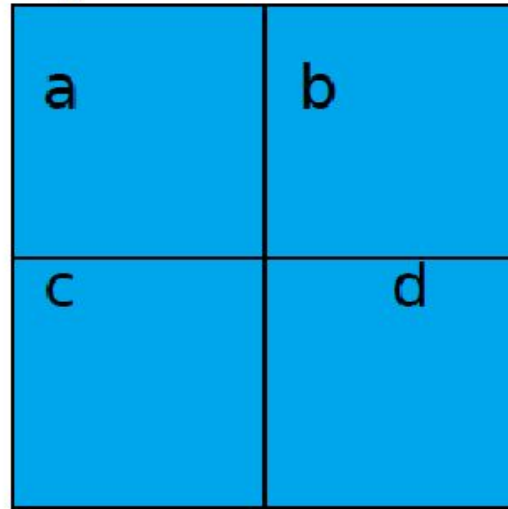
$a+b$
$c+d$

a
$c+b$
d

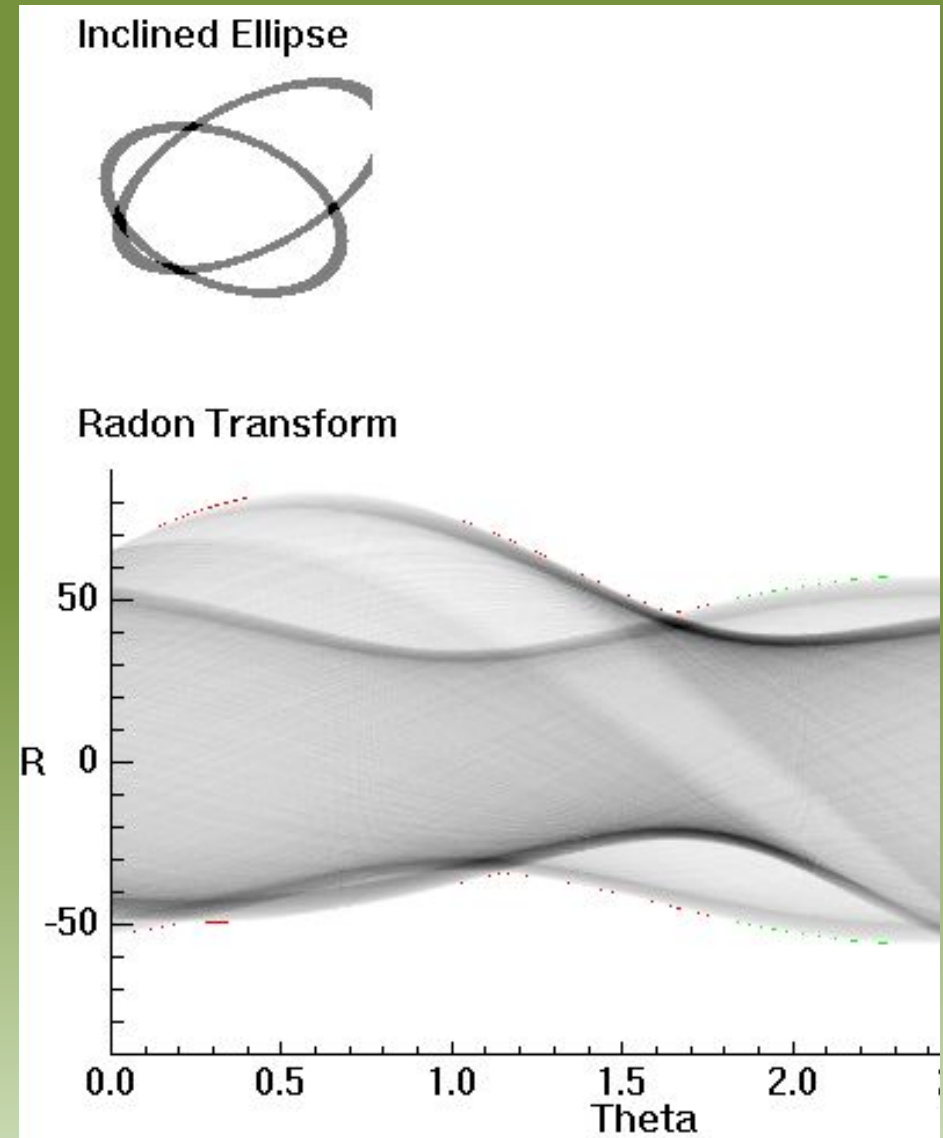
$a+c$	$b+d$
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b
$a+d$
c

Divide by number of projections $-1 = 3$



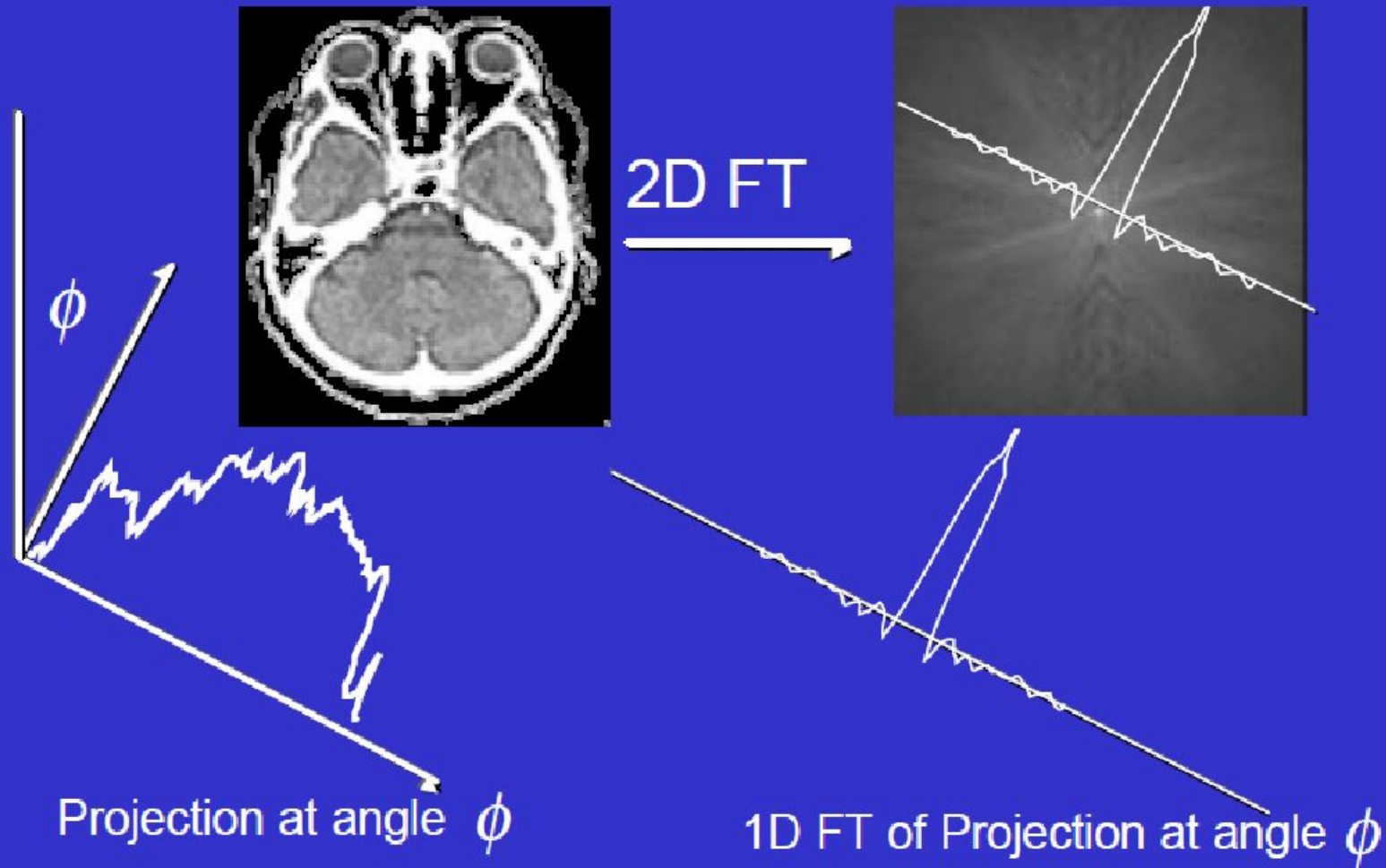
Sinogram



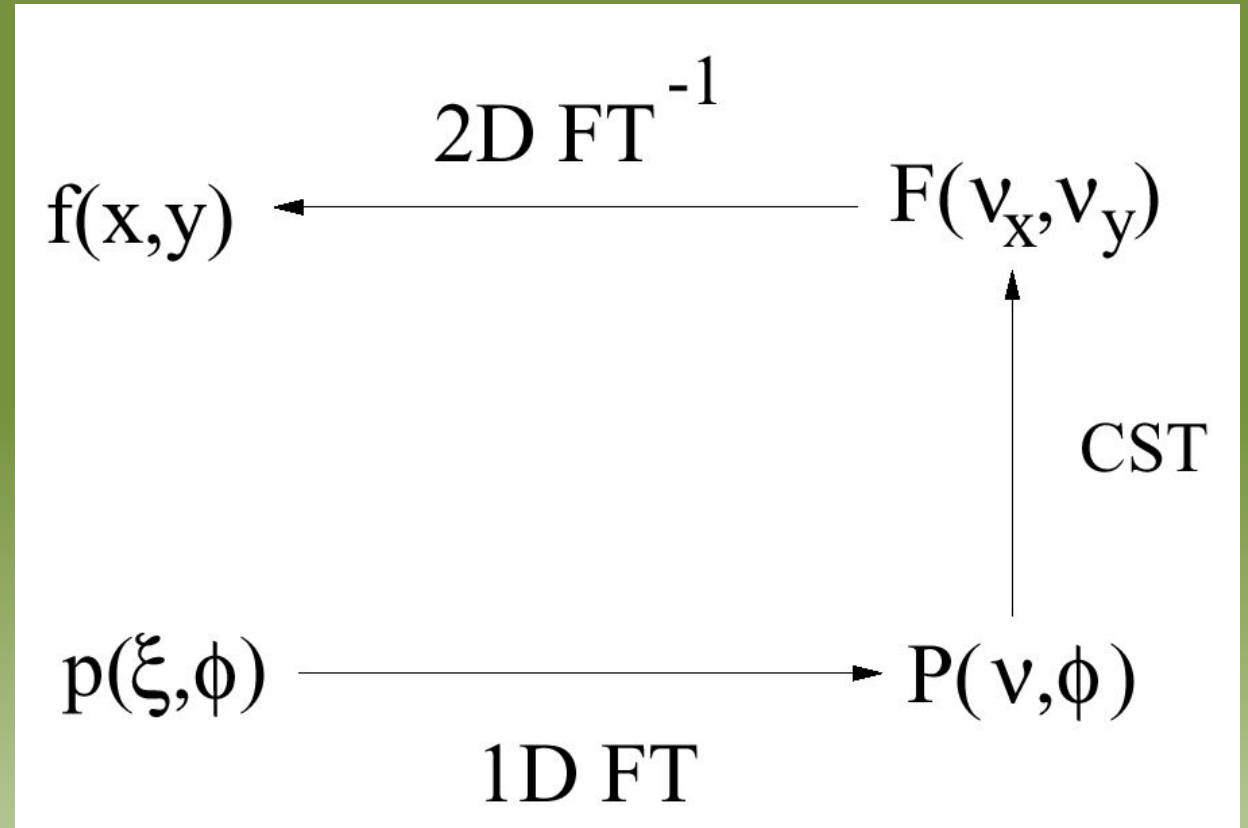
Central Slice Theorem

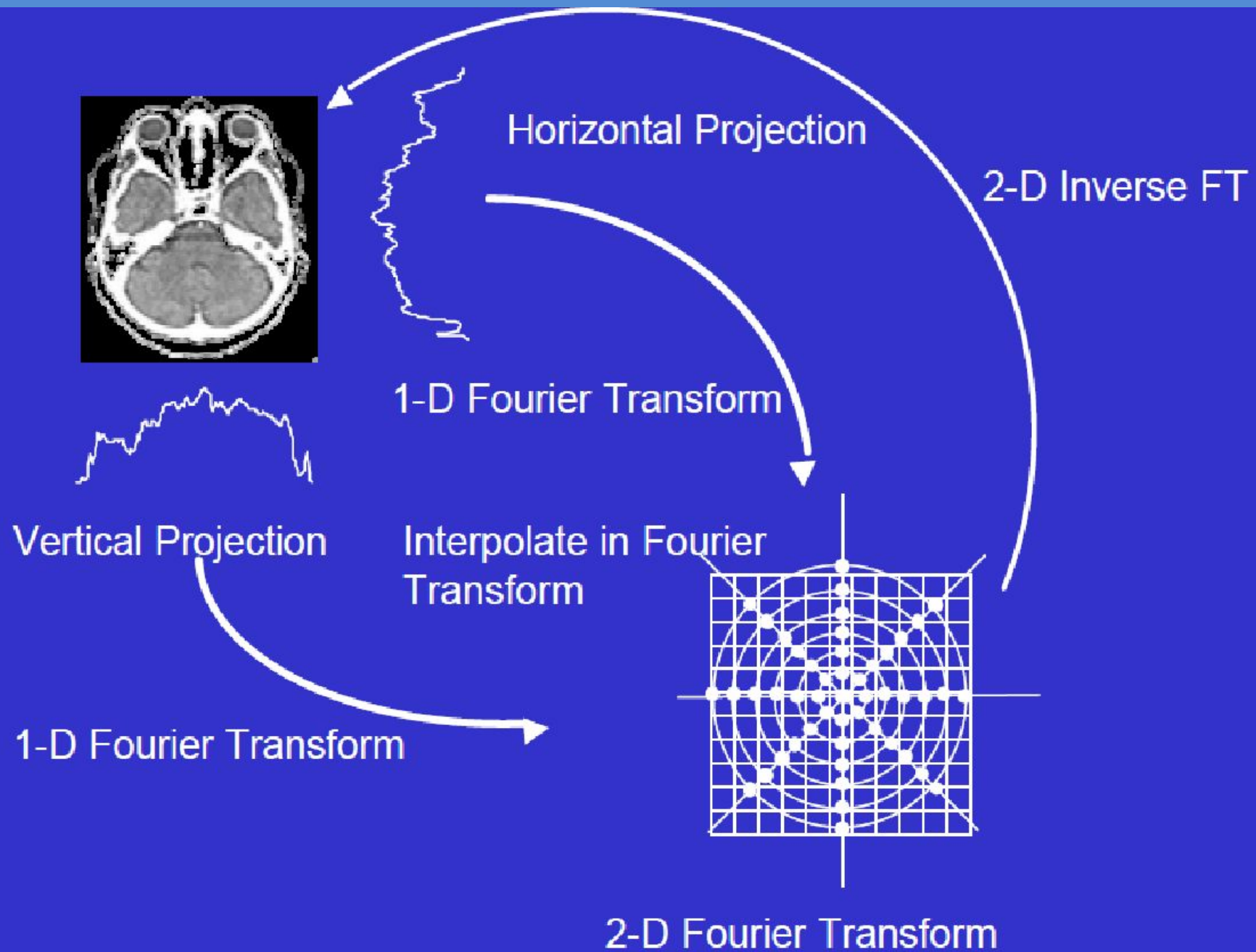
- Pivotal to understanding of CT reconstruction
- Relates 2D FT of image to 1D FT of its projection
- N.B. 2D FT is “k-space” of MRI

Central Slice Theorem



- Calculate the 1D Fourier transform of all projections [$p(r) = P(k)$]
- Place $P(k)$ on polar grid to get $P(k, \Theta)$
- Resample in Cartesian space to get $F(k_x, k_y)$
- Calculate the 2D inverse Fourier transform of $F(k_x, k_y)$ to get $f(x, y)$ – image
- Resultant image is noisy





Filtered Back-projection

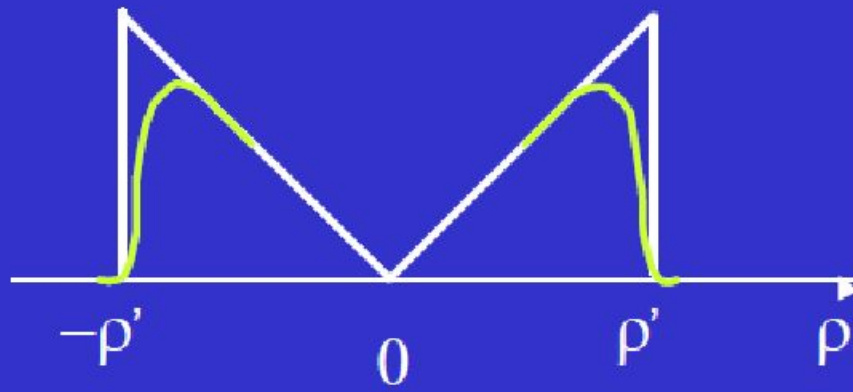
- Direct back-projection results in blurred image
- Could filter (de-convolve) resulting 2-D image
- Linear systems theory suggests order of operations unimportant
- Filter projection profiles **prior** to back projection

Convolution Filter

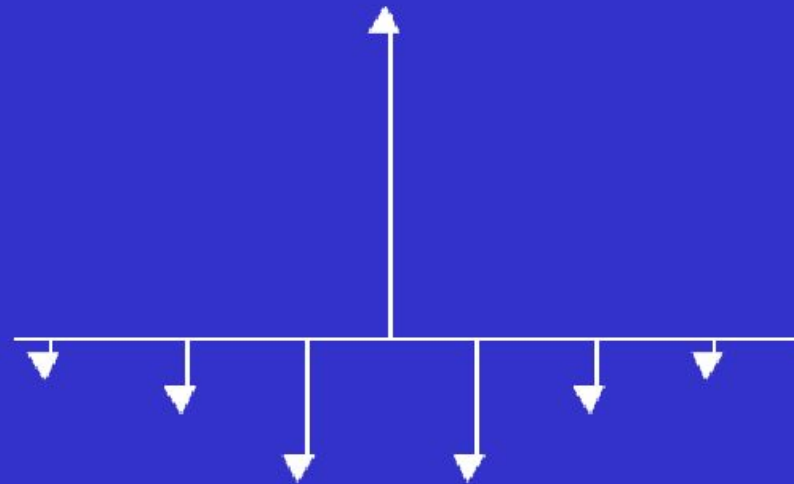
In Fourier Space, filter has the form $|\rho|$

Maximum frequency must be truncated at ρ'

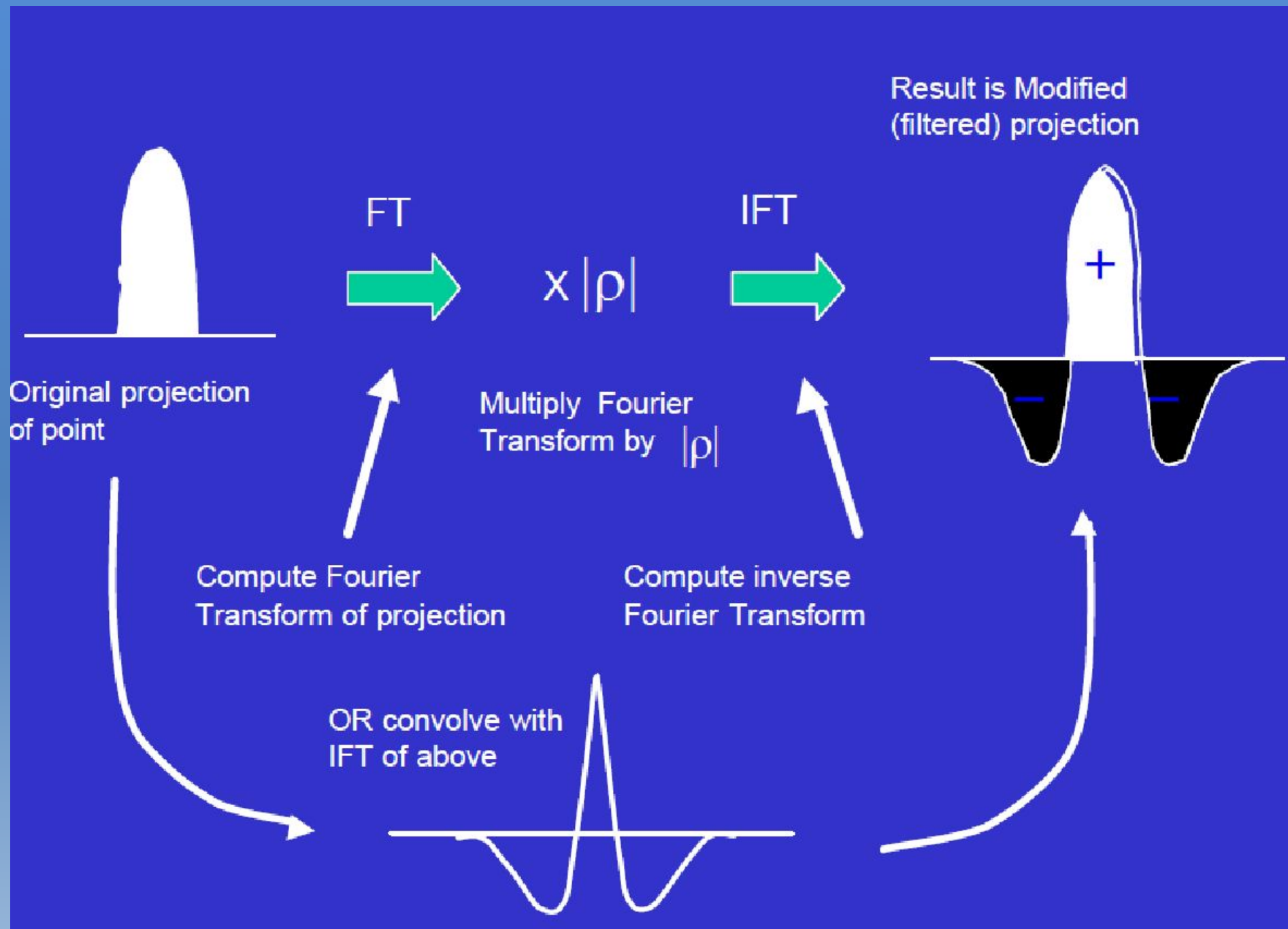
Filtering may occur in Fourier or Spatial domain



Frequency response of filter



Sampled convolution filter



Back-projecting the Filtered Projection





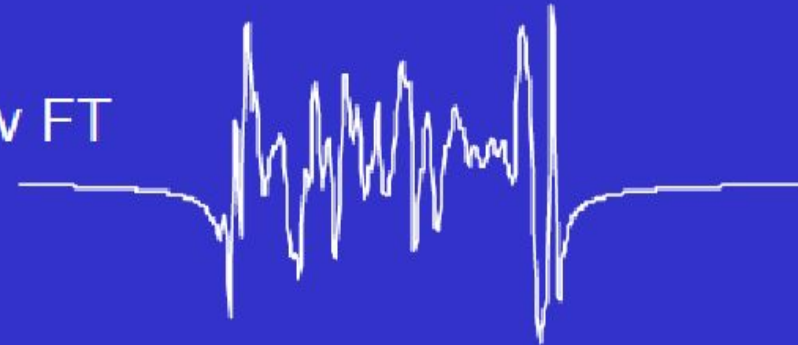
Cross-section of head

Back-project filtered
projections (at all angles)



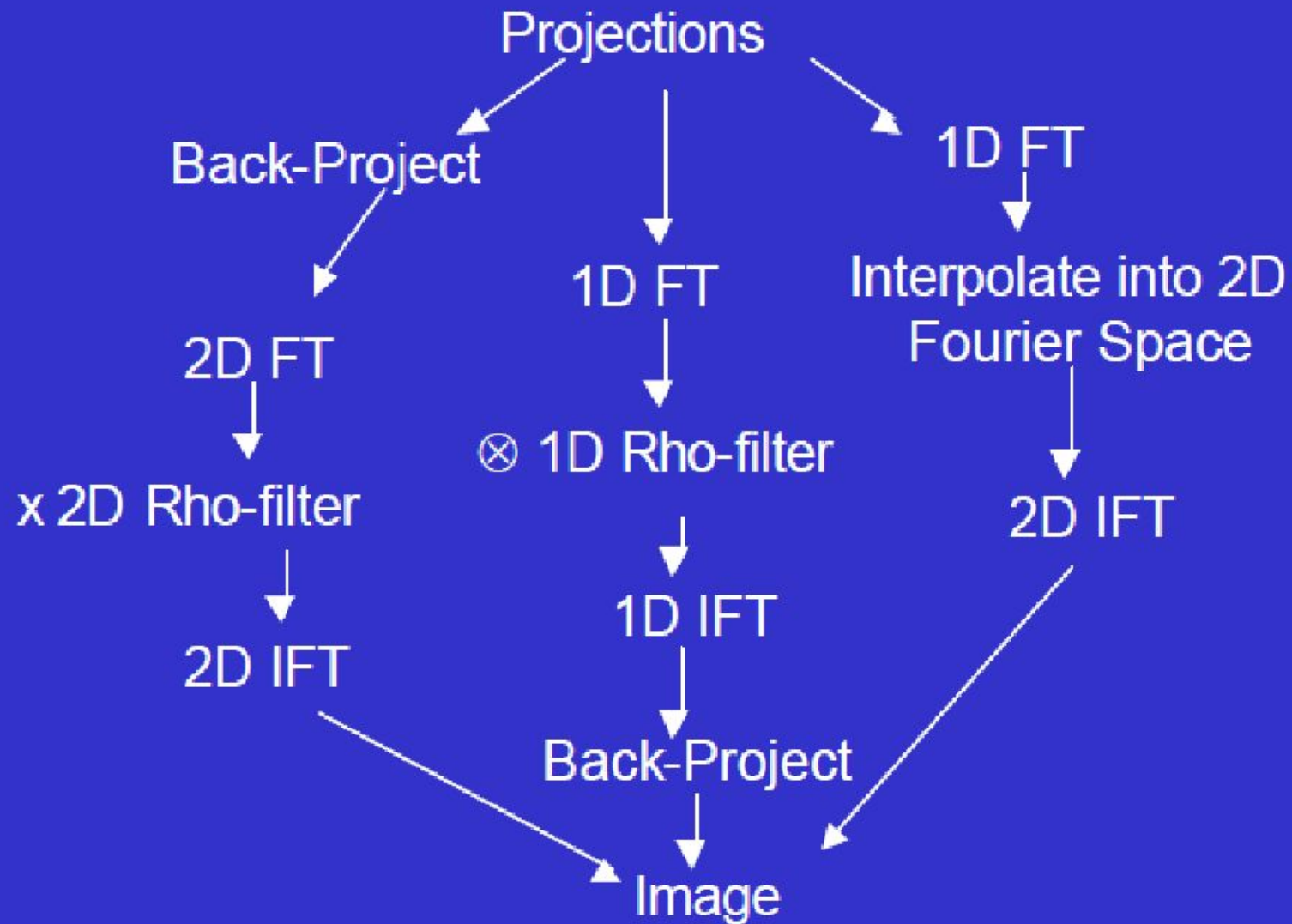
Vertical projection
of this cross-section

FT $\rightarrow$ $|\rho|$ $\rightarrow$ Inv FT

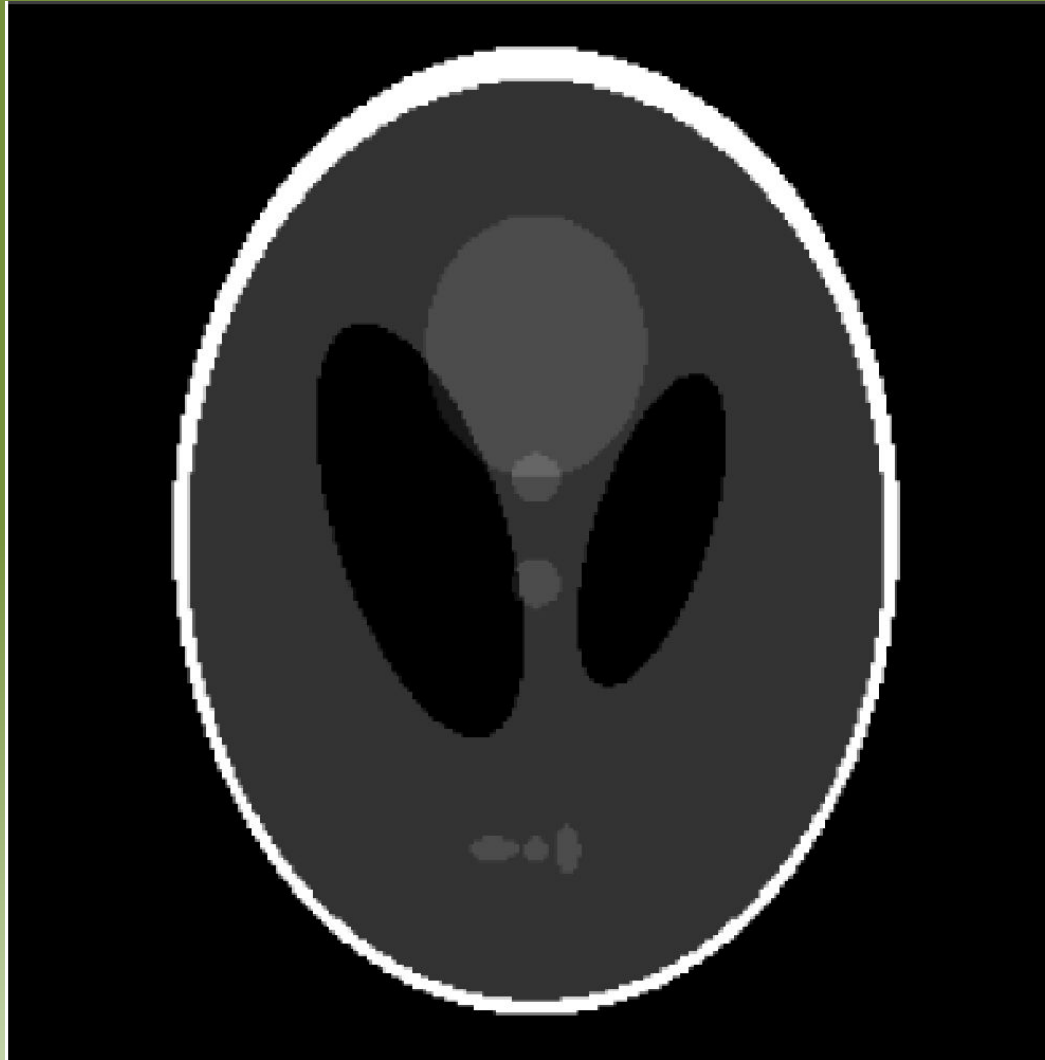


Modified (filtered) projection

Projections to Images



Shep & Logan Phantom



Assignment

Q1

A sinogram is a 2D plot showing all the projections for a CT scan. On the horizontal axis, each projection is displayed. Each vertical line has the data from a different projections. The vertical lines are organized so that the projection angle increases from 0 to 180 degrees. A point in the image will move as a sine wave in a sinogram – hence the name.

Generate the sonogram of a Shepp Logan phantom

Clues

Decide number of projections and use Radon transfrms (say eg. projections with an angular spacing of 0.5° over 180° . Hint: you can use the MATLAB function *radon*. Provide an image of the sinogram.

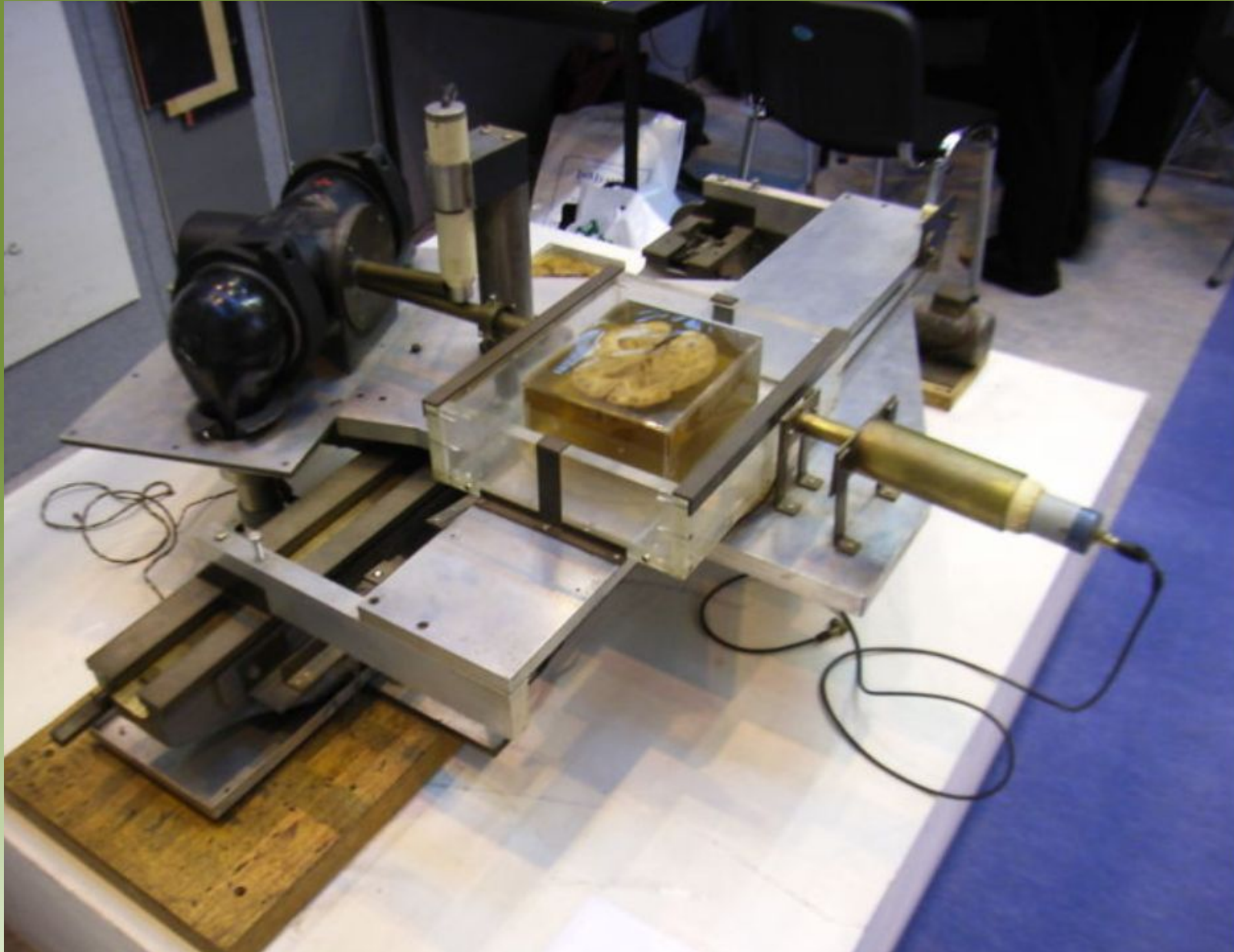
Q2: Simulate the reconstruction of a CT scanner as the scanner acquires projections and transforms these projections into images. You can use your own code to backproject or download any code (numerous matlab version of bp.m, backproj.m, bproject.m Fbproj.m available on net.

Explain what you did (understand the code and explain the results.

Q3. Use a good filter (eg. Mexican hat wavelet) or any other along with back projection and redo the exercise and show the results of filtered backprojection.

Compare the two results and comment how you can make it better.

FIRST CT SCANNER PROTOTYPE

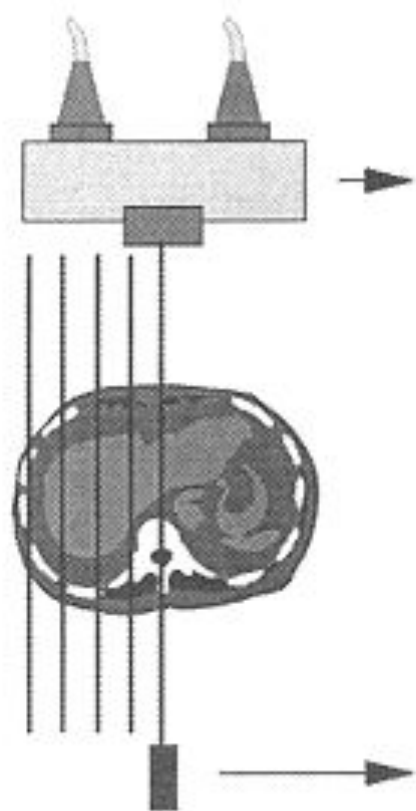


1st generation: rotate/translate, pencil beam

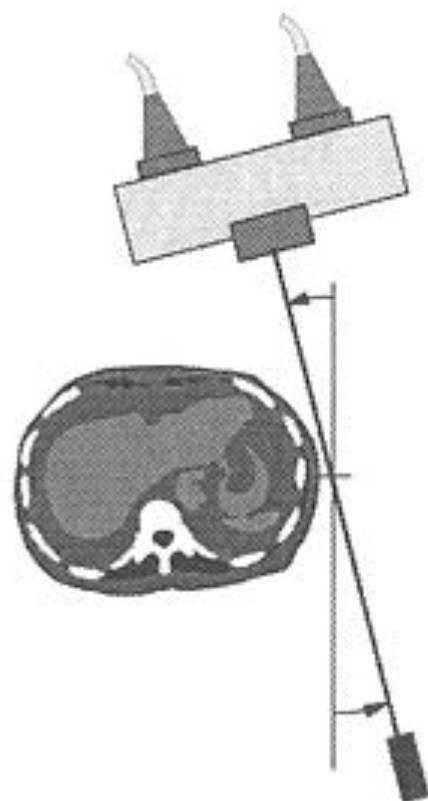
- Only 2 x-ray detectors used (two different slices)
- Parallel ray geometry
- Translated linearly to acquire 160 rays across a 24 cm FOV
- Rotated slightly between translations to acquire 180 projections at 1-degree intervals
- About 4.5 minutes/scan with 1.5 minutes to reconstruct slice

1st generation (cont.)

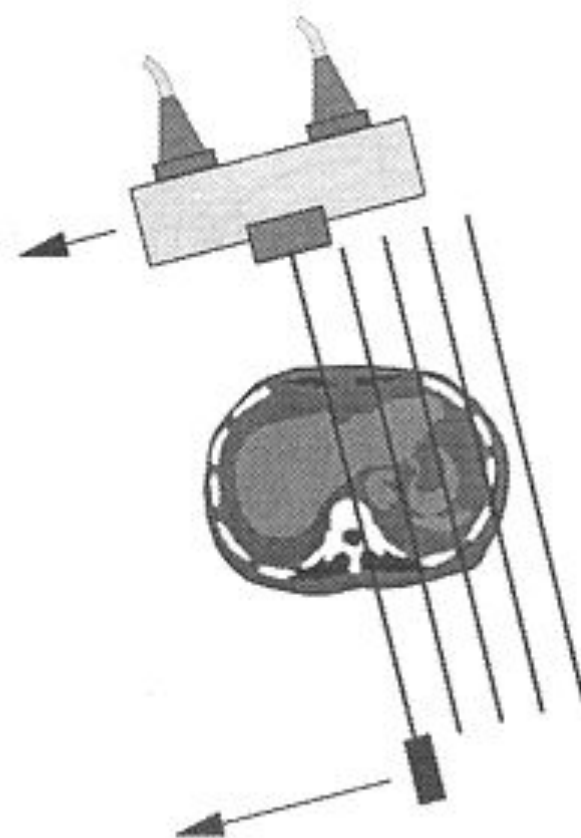
- Large change in signal due to increased x-ray flux outside of head
 - Solved by pressing patient's head into a flexible membrane surrounded by a water bath
- NaI detector signal decayed slowly, affecting measurements made temporally too close together
- Pencil beam geometry allowed very efficient scatter reduction, best of all scanner generations



translate



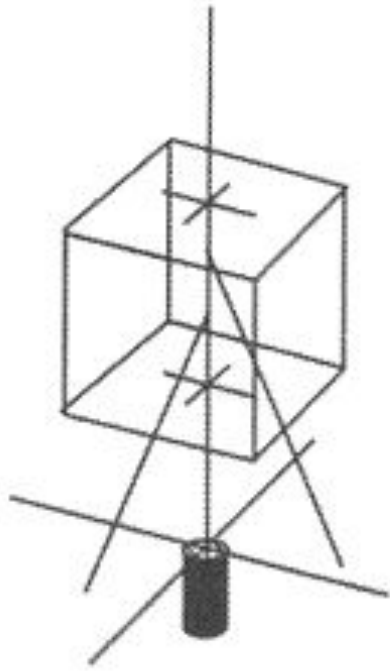
rotate



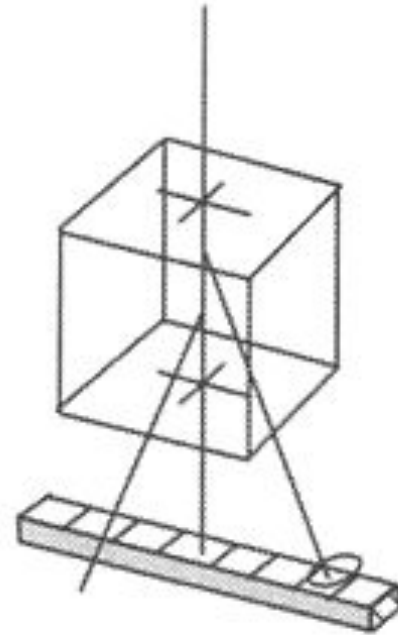
translate

2nd generation: rotate/translate, narrow fan beam

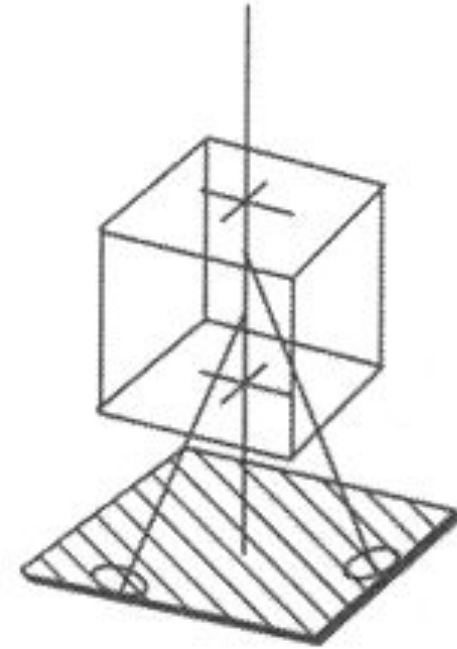
- Incorporated linear array of 30 detectors
- More data acquired to improve image quality (600 rays x 540 views)
- Shortest scan time was 18 seconds/slice
- Narrow fan beam allows more scattered radiation to be detected



Pencil Beam
Geometry



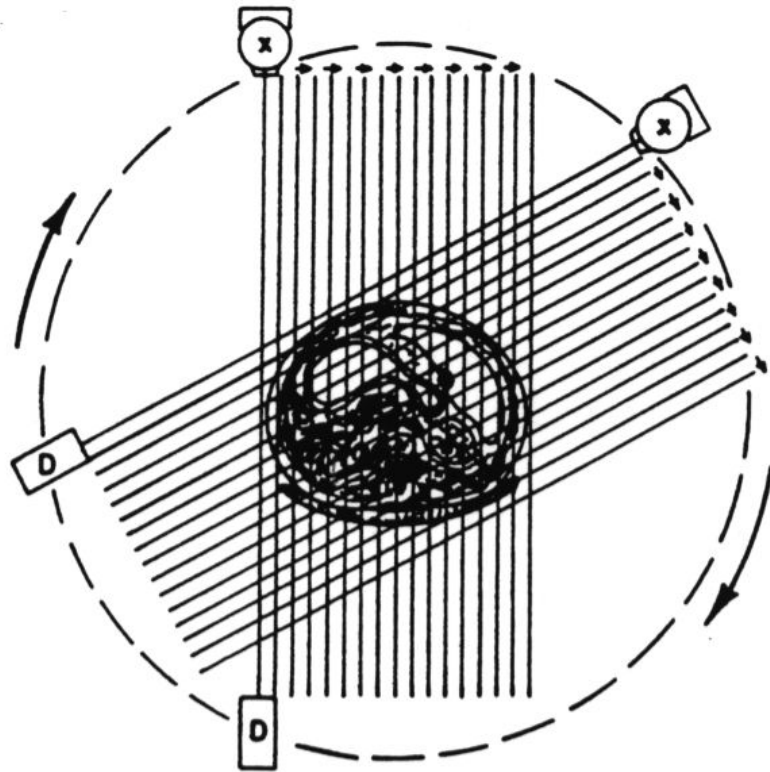
Fan Beam
Geometry



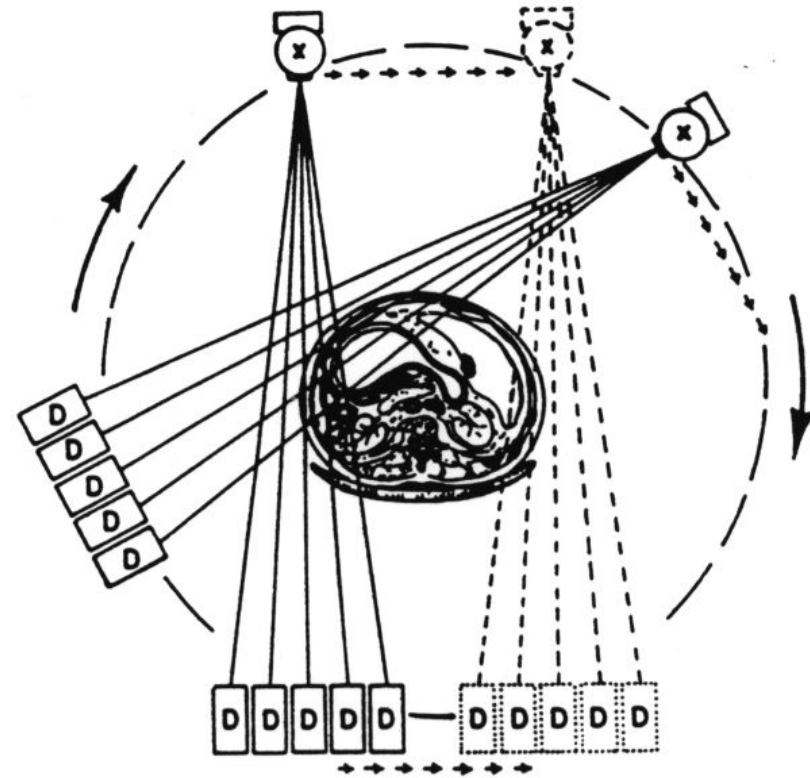
Open Beam
Geometry

CT “generations”

1st generation



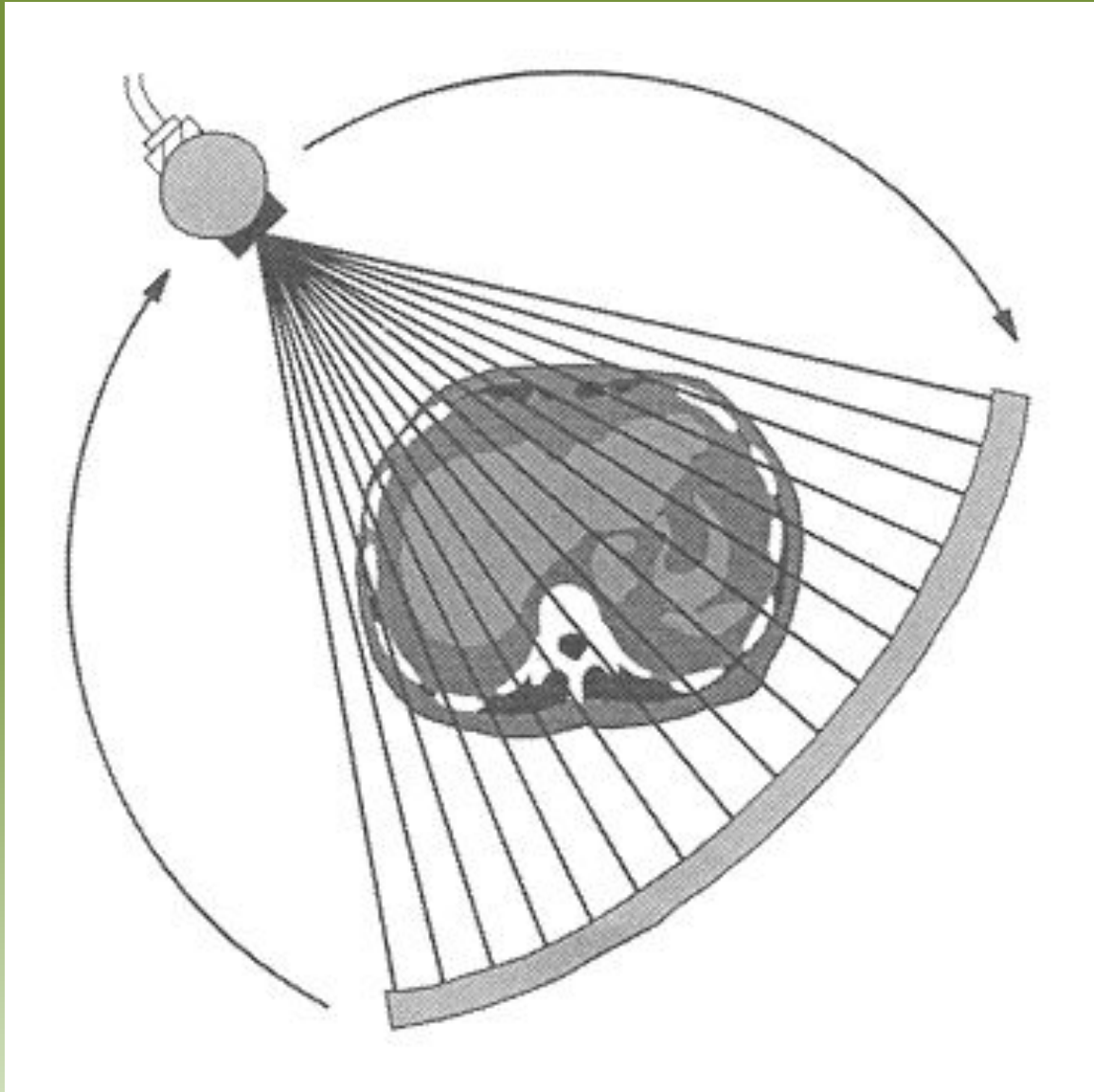
2nd generation



translation-rotation scanners

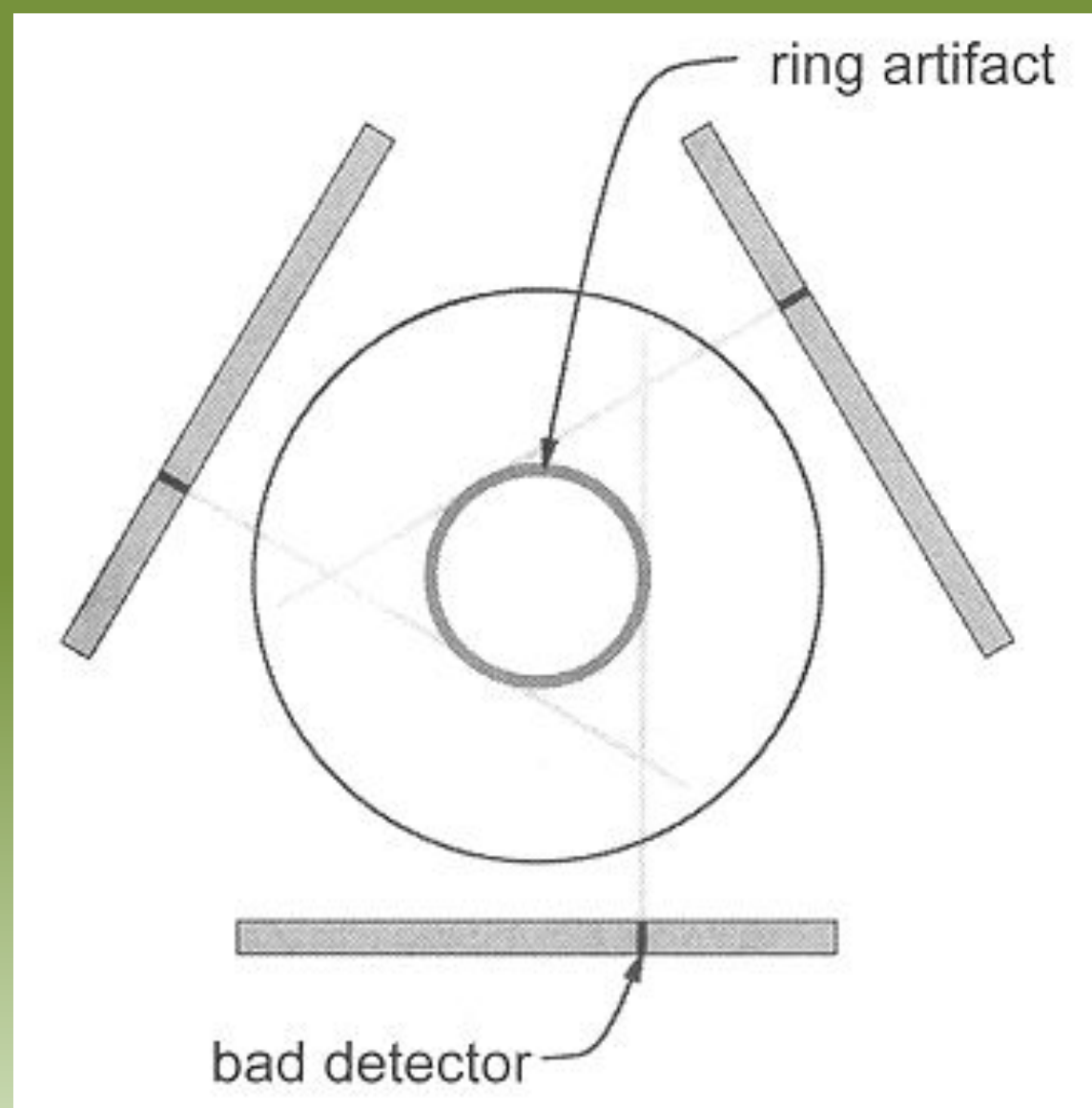
3rd generation: rotate/rotate, wide fan beam

- Number of detectors increased substantially (to more than 800 detectors)
- Angle of fan beam increased to cover entire patient
 - Eliminated need for translational motion
- Mechanically joined x-ray tube and detector array rotate together
- Newer systems have scan times of ½ second



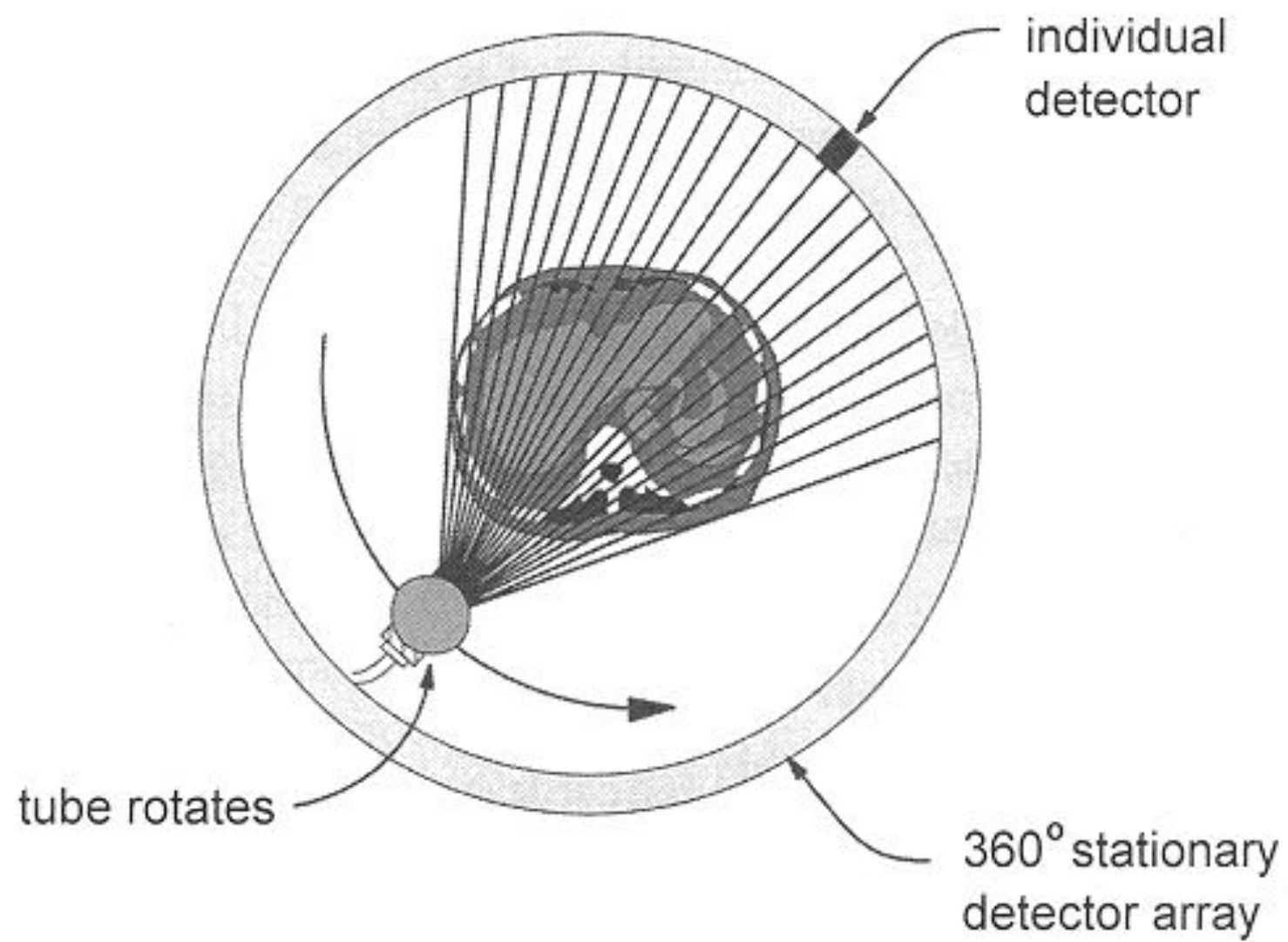
Ring artifacts

- The rotate/rotate geometry of 3rd generation scanners leads to a situation in which each detector is responsible for the data corresponding to a ring in the image
- Drift in the signal levels of the detectors over time affects the μ t values that are backprojected to produce the CT image, causing ring artifacts



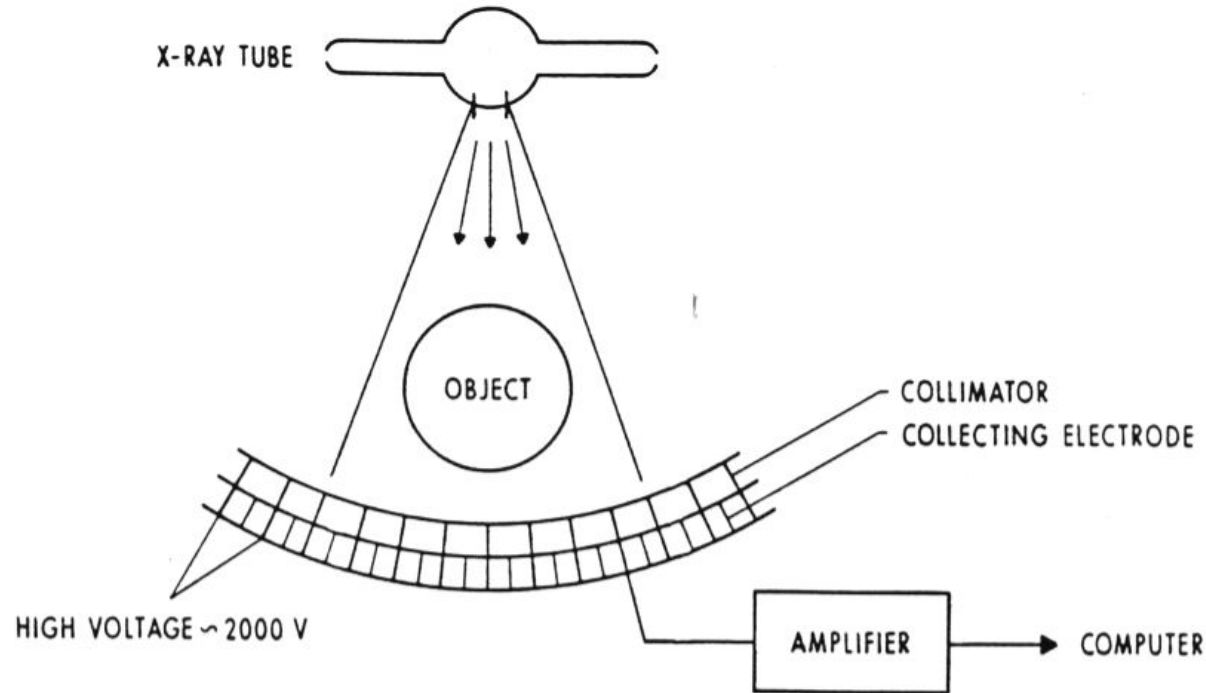
4th generation: rotate/stationary

- Designed to overcome the problem of ring artifacts
- Stationary ring of about 4,800 detectors

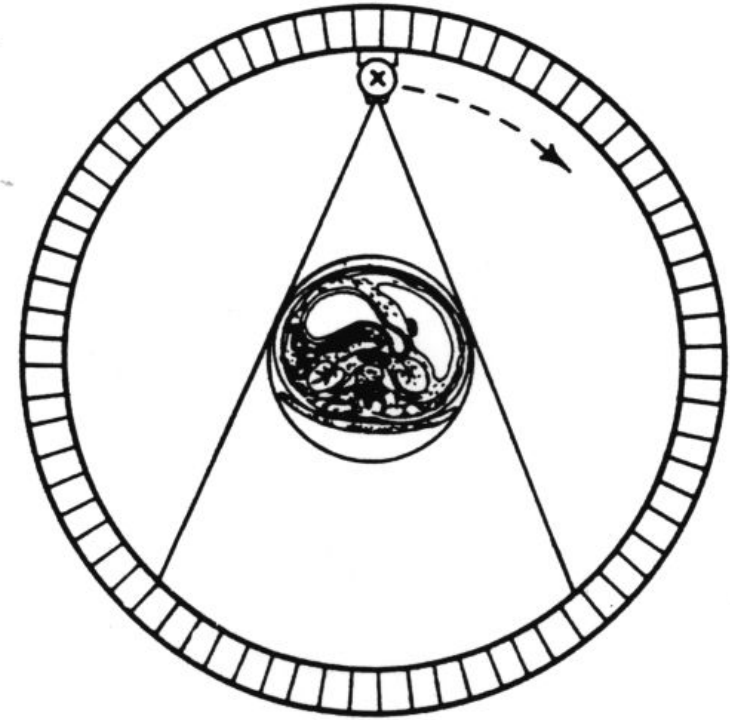


CT “generations”

3rd generation
(fan detector)



4th generation
(ring detector)



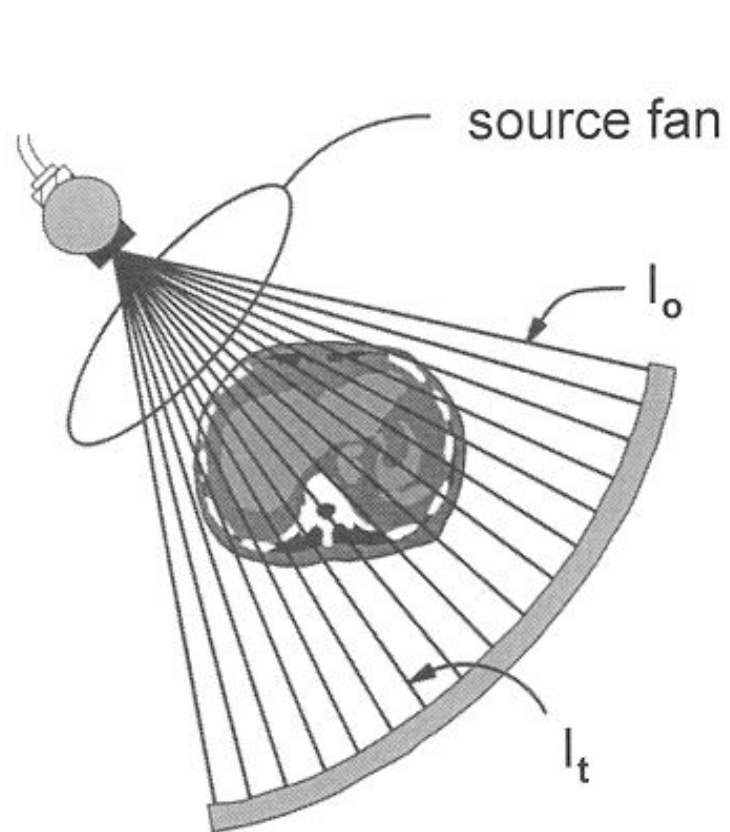
Rotation-only scanners

3rd vs. 4th generation

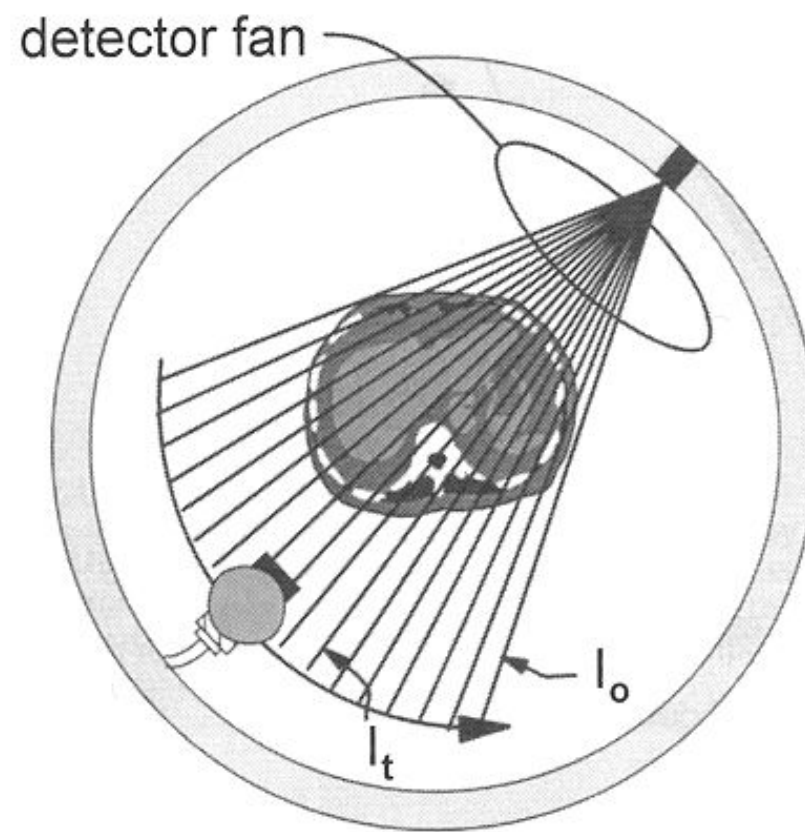
- 3rd generation fan beam geometry has the x-ray tube as the apex of the fan; 4th generation has the individual detector as the apex

$$3^{\text{rd}} \text{ gen} : \ln(g_1 I_0 / g_2 I_t) = \mu t$$

$$4^{\text{th}} \text{ gen} : \ln(g I_0 / g I_t) = \mu t$$



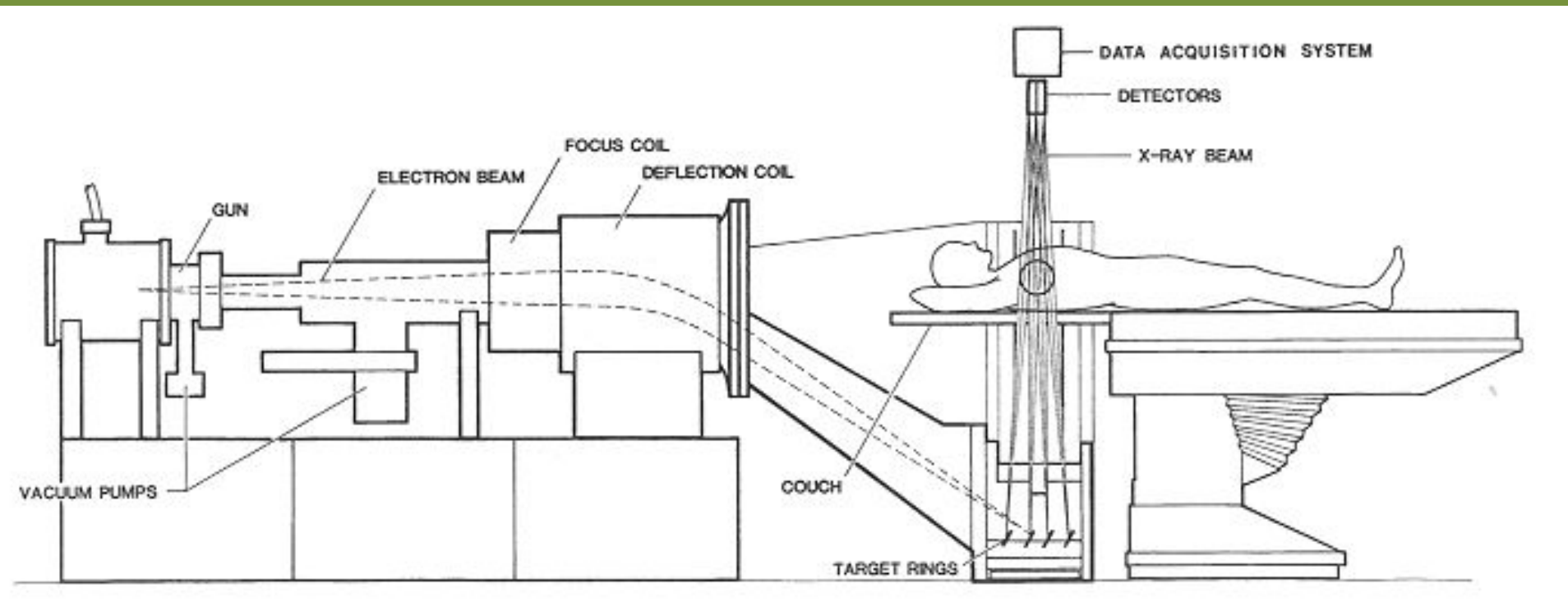
third generation



fourth generation

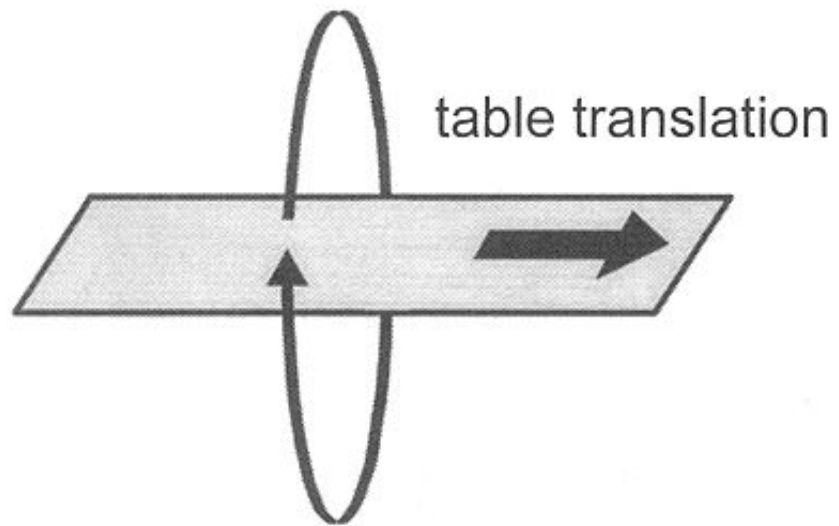
5th generation: stationary/stationary

- Developed specifically for cardiac tomographic imaging
- No conventional x-ray tube; large arc of tungsten encircles patient and lies directly opposite to the detector ring
- Electron beam steered around the patient to strike the annular tungsten target
- Capable of 50-msec scan times; can produce fast-frame-rate CT movies of the beating heart

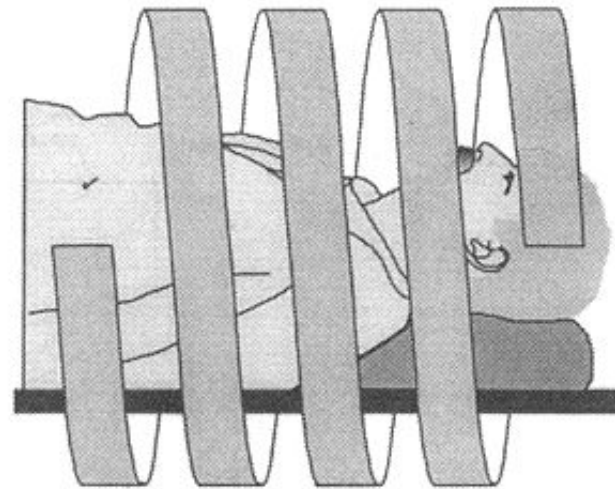


6th generation: helical

- Helical CT scanners acquire data while the table is moving
- By avoiding the time required to translate the patient table, the total scan time required to image the patient can be much shorter
- Allows the use of less contrast agent and increases patient throughput
- In some instances the entire scan can be done within a single breath-hold of the patient



x-ray tube rotation



helical x-ray tube
path around patient

7th generation: multiple detector array

- When using multiple detector arrays, the collimator spacing is wider and more of the x-rays that are produced by the tube are used in producing image data
 - Opening up the collimator in a single array scanner increases the slice thickness, reducing spatial resolution in the slice thickness dimension
 - With multiple detector array scanners, slice thickness is determined by detector size, not by the collimator

